



THE UNIVERSITY OF
MELBOURNE

Probability Reliability and Quality

Reliability and Maintenance

Kevin Otto

Schedule and Learning Outcomes

Week	Recorded Lecture Topic
1	Introduction.
2	Inspection. Yield.
3	ANOVA. Measurement Systems Analysis.
4	Statistics. Confidence Intervals.
5	Hypothesis Testing. Sample Sizes.
6	Quality Management Processes.
7	Regression.
8	Design of Experiments.
9	System Reliability. Availability.
	<i>BREAK – Non Teaching Week</i>
10	Parametric Distribution Analysis
11	Statistical Process Control. Supplier Quality.
12	Review

After completion of the course the student should be able to:

- Explain different statistical methods
- Parametric and non-parametric hypothesis tests
- Design and plan for collecting experimental data
- **Concepts of reliability engineering and the relationship between TTF, R, H**
- How to improve its processes, products and services
- Apply TQM, 6S, and benchmarking



Overview

Reliability

Weibull distribution

System Reliability

MTBF

Hazard Rate

Maintenance Plans



Definition of Reliability

Overall, a product's *reliability* is:

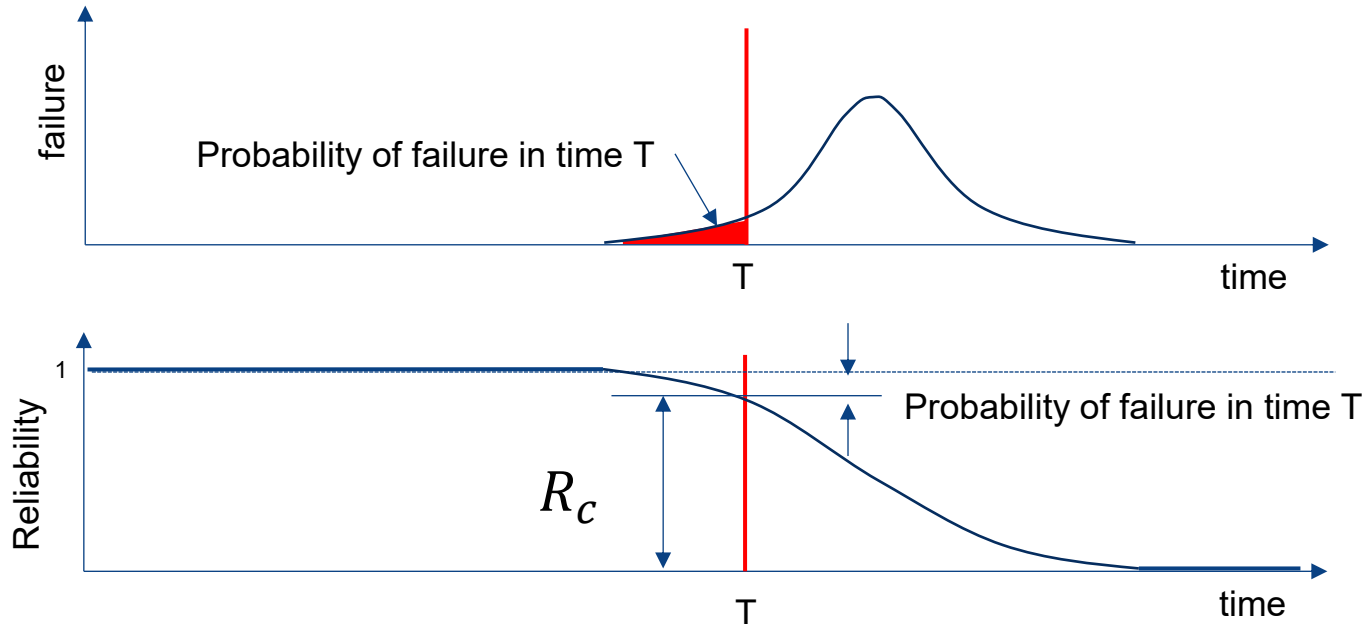
the probability that a product will operate within specification for a specified duration under the design operating conditions without failure.

- Probability. Nothing is certain into the future.
- Will operate. It meets all specifications. It makes the customer happy.
- Without failure. Specified duration. Specified operating conditions.

Reliability

Probability-of-failure and reliability of a component are distribution functions of time.

If we define a time period T then we can define the **reliability** R_c of a component.





Time and Cycle Based Reliability

A component may operate constantly, like a motor or engine.

A component may operate for instants, like flipping a switch.

A part in a system may be used 100% of when the system operates, a battery.

A part in a system may be in standby mode part of the time, a brake.

We use both in a system rollup of reliability.

We need to include the *utilization* rate of every part.



Cycles and Utilization

If the part is actuated at a rate r then what is its equivalent life?

The life L (years) is related to the rate r (actuations/yr) and number of actuations N :

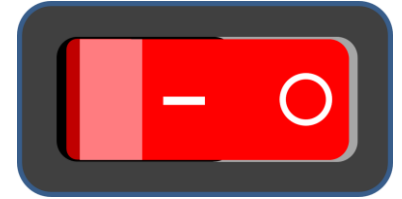
$$L = N/r$$

Cycles and Utilization

A switch is rated for $N = 10,000$ cycles.

How many years will it last?

The utilization rate r is also needed to compute life in years.



If the switch is actuated at $r = 4$ times per day, how many years is rated to last?

Rated life (years) : $L = N/r$

$$\text{Switch: } L = \frac{N}{r} = \frac{10,000}{4 \frac{365}{1}} = 6.85 \text{ years}$$



Key Reliability Metrics

Probability density function

(failures)

Cumulative distribution function

(cumulative failure)

Reliability function

(cumulative survival)

Hazard function

(failure rate)

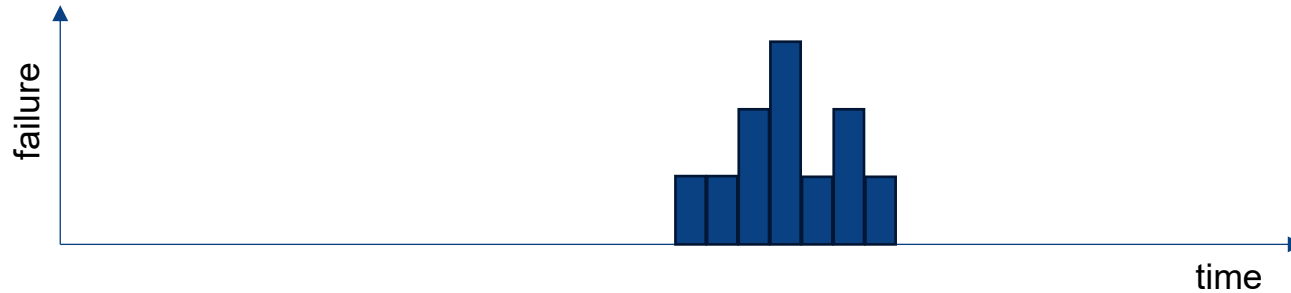
Basic Reliability Functions

The analysis can also be completed using the histogram of counts of failures.

- The histogram of failures over time is analogous to the probability density function

All the other relationships can be derived from the histogram of failures

- Cumulative failures
- Survived
- Failure rate



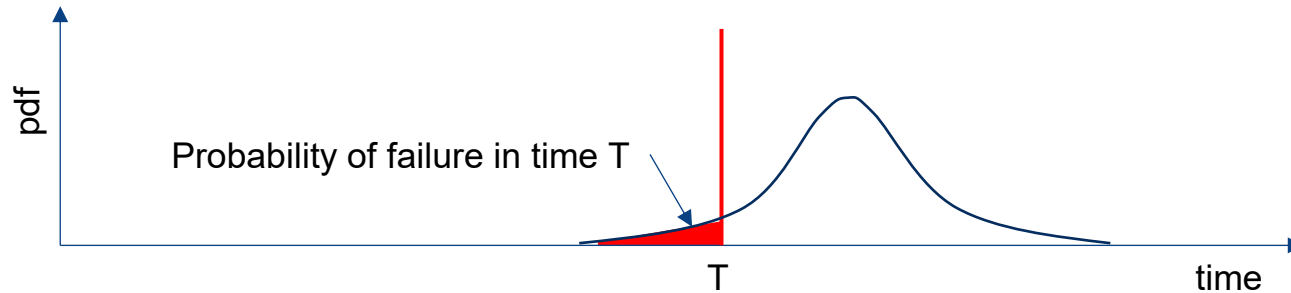
Basic Reliability Functions

The ***probability density function*** is an important reliability function

- The histogram of failures over time normalized to have an integral of 1.

All the other relationships can be derived from the density function

- Cumulative distribution function
- Reliability function
- Hazard rate function



Example

Suppose we run a life test on a component and accumulate 50 failures over a testing period of 5000 hrs.

OR: Suppose we run a process and accumulate 50 shutdown failures over a testing period of 5000 hrs.

The failures f_i are put into "bins" of 1000 hrs and appear as follows

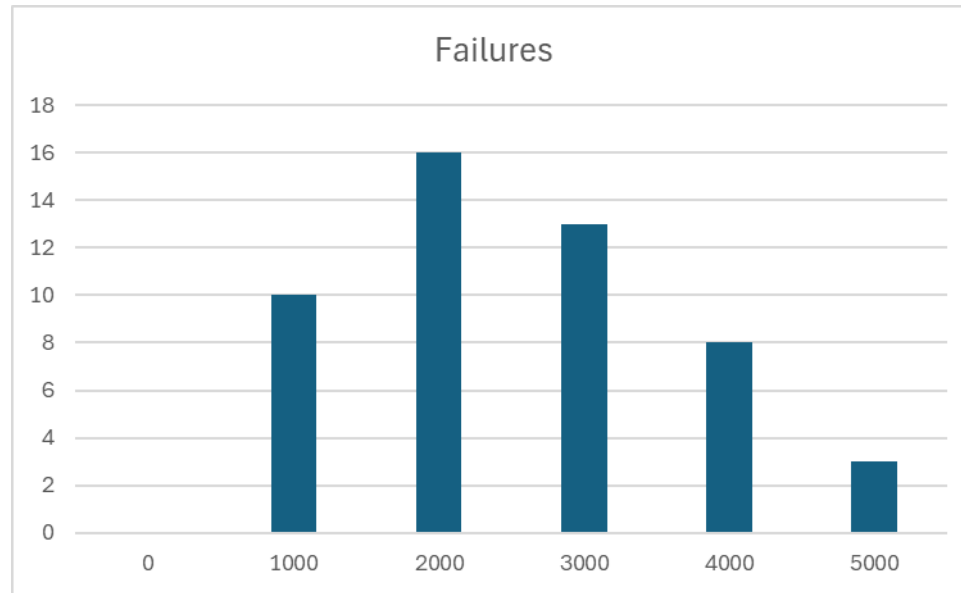
Hours	Failures
0-1000	10
1000-2000	16
2000-3000	13
3000-4000	8
4000-5000	3
TOTAL	50

Failures Histogram

Graphing the failures f_i show times when more and less fail.

f

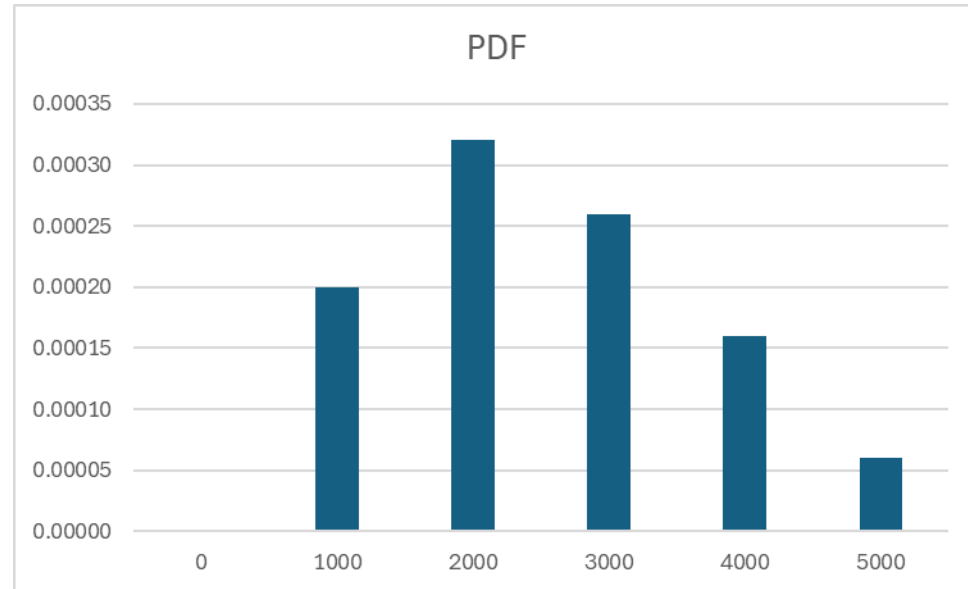
Hours	Failures
0	0
1000	10
2000	16
3000	13
4000	8
5000	3



Probability Density Function

The life ***probability density function*** is the percent failures divided by the bin width in time (hours).
$$PDF_i = \frac{\# \text{ failed in interval } i}{\text{Total \#} * \text{time interval}} = f_i / (N\tau)$$

<i>f</i>		<i>PDF</i>
Hours	Failures	PDF
0	0	0.0000
1000	10	0.00020
2000	16	0.00032
3000	13	0.00026
4000	8	0.00016
5000	3	0.00006



Cumulative Failed

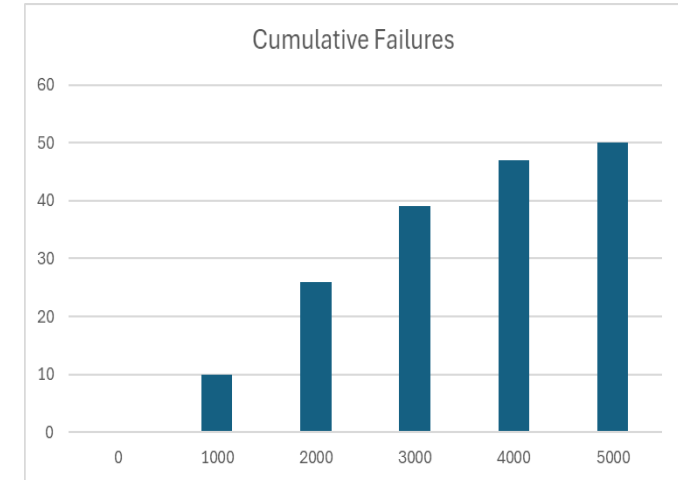
The ***cumulative failed*** is the sum of those that have failed by a certain time.

For any histogram bin,
the cumulative failed is

$Failed_i = \text{\#failed up to period } i$

$$C_i = \sum_{k=1}^i f_k$$

	f	C
Hours	Failures	Failed
0	0	0
1000	10	10
2000	16	26
3000	13	39
4000	8	47
5000	3	50



Cumulative Failure Distribution or Unreliability

The ***cumulative failure distribution*** is the fraction of the population that has failed by a certain time. Also called the *cumulative distribution function* CDF.

For any histogram bin,
the cumulative failure rate is

$$CDF_i = \frac{\text{\#failed up to period } i}{\text{Total}}$$

$$= \frac{\sum_{k=1}^i f_k}{N}$$

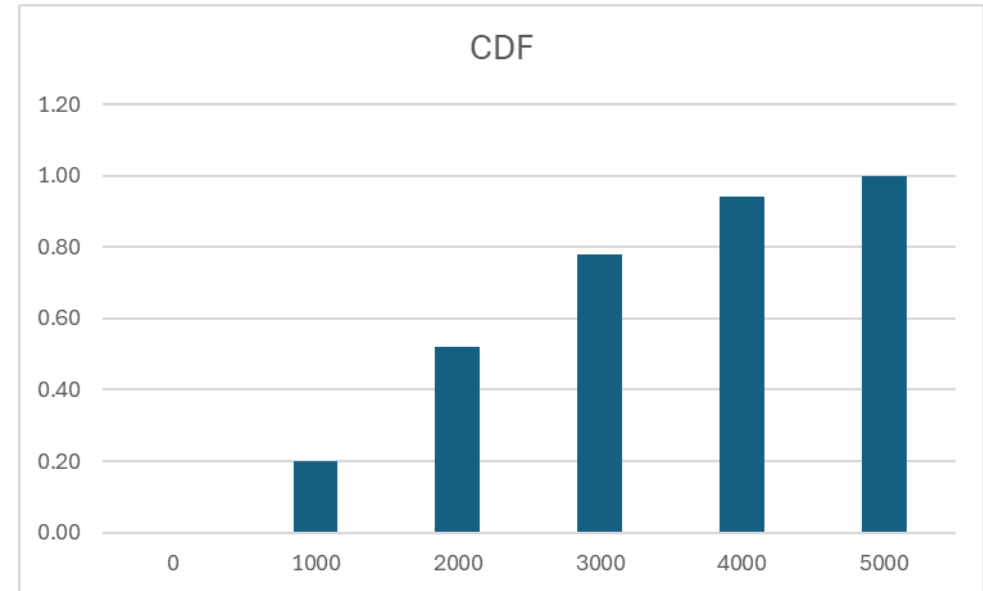
	<i>f</i>	<i>C</i>	<i>PDF</i>	<i>CDF</i>
Hours	Failures	Failed	PDF	CDF
0	0	0	0.00000	0.00
1000	10	10	0.00020	0.20
2000	16	26	0.00032	0.52
3000	13	39	0.00026	0.78
4000	8	47	0.00016	0.94
5000	3	50	0.00006	1.00

It is also equivalently $CDF_i = PDF_i * \tau + CDF_{i-1}$

It is also equivalently $CDF_i = C_i/N$

Fraction Failed – Cumulative Distribution

	f	C	PDF	CDF
Hours	Failures	Failed	PDF	CDF
0	0	0	0.0000	0.00
1000	10	10	0.00020	0.20
2000	16	26	0.00032	0.52
3000	13	39	0.00026	0.78
4000	8	47	0.00016	0.94
5000	3	50	0.00006	1.00

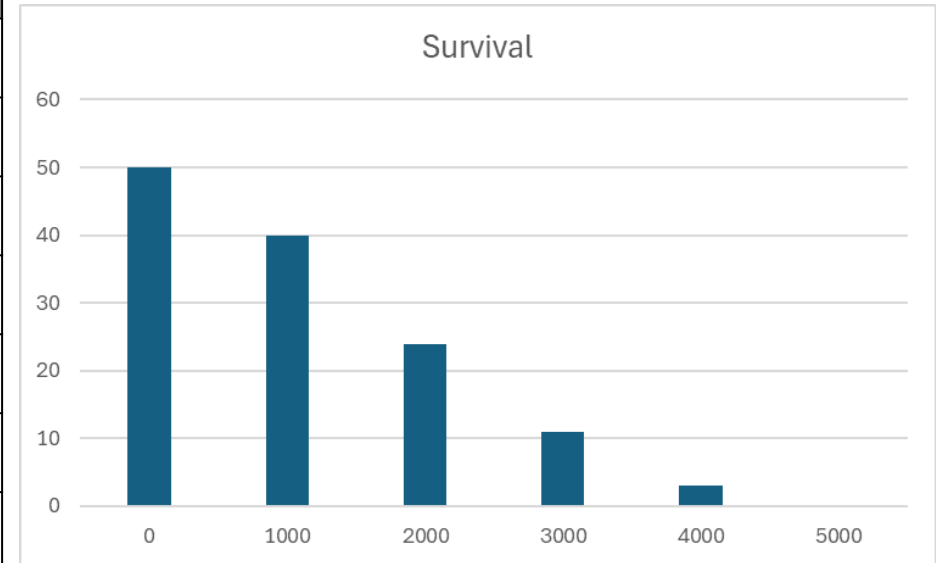


Survival Function

Next consider how many have survived at each interval, the **survival**.

	f	C	S
Hours	Failures	Failed	Survived (at start)
0	0	0	50
1000	10	10	40
2000	16	26	24
3000	13	39	11
4000	8	47	3
5000	3	50	0
TOTAL	50		

S_i = number survived at start of time i
 $S_i = N - C_i$



The **survival function** or the **reliability distribution** is the complement to the cumulative life distribution.

For any histogram bin,
the reliability is

$$R_i = 1 - CDF_i$$

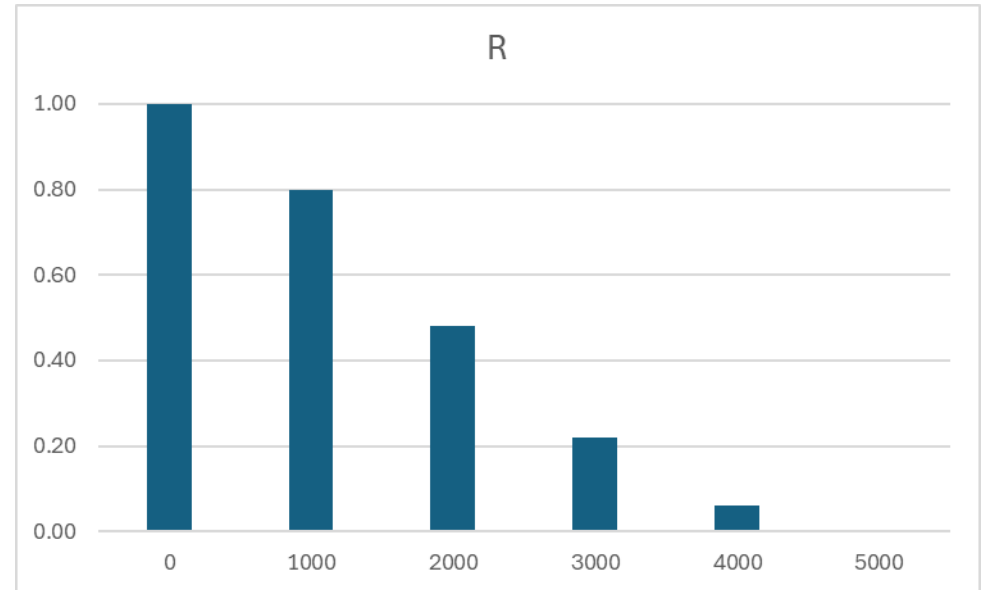
It is also equivalently

$$R_i = S_i/N$$

	f	C	S	PDF	CDF	R
Hours	Failures	Failed	Survived (at start)	PDF	CDF	Rel
0	0	0	50	0.0000	0.00	1.00
1000	10	10	40	0.00020	0.20	0.80
2000	16	26	24	0.00032	0.52	0.48
3000	13	39	11	0.00026	0.78	0.22
4000	8	47	3	0.00016	0.94	0.06
5000	3	50	0	0.00006	1.00	0.00

Reliability

	<i>PDF</i>	<i>CDF</i>	<i>R</i>
Hours	PDF	CDF	Rel
0	0.0000	0.00	1.00
1000	0.00020	0.20	0.80
2000	0.00032	0.52	0.48
3000	0.00026	0.78	0.22
4000	0.00016	0.94	0.06
5000	0.00006	1.00	0.00





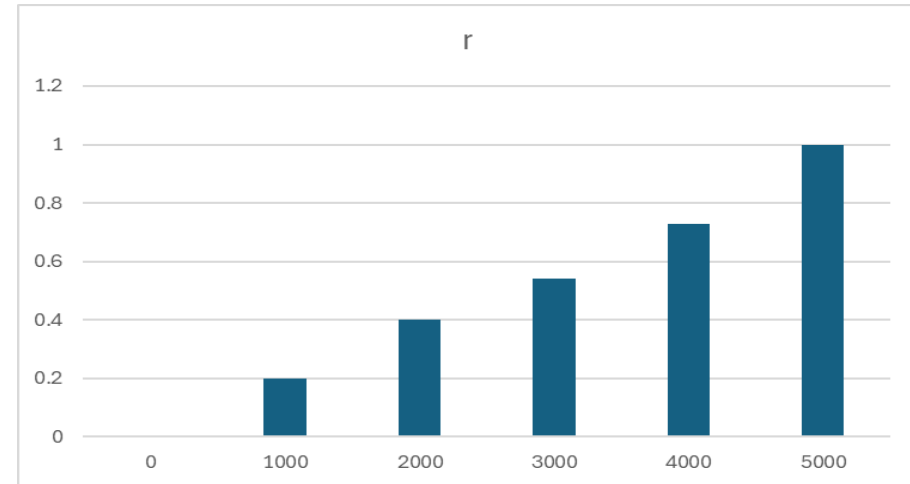
Failure Rate

The **failure rate** in an incremental time period is the fraction that fail of those that survived up until then.

For any histogram bin, the failure rate is

$$r_i = \frac{\text{\#failed period } i}{\text{\# survived start period } i} = \frac{f_i}{S_{i-1}}$$

	f	C	S	r
Hours	Failures	Failed	Survived (Period Start)	Failure Rate
0	0	0	50	0.00
1000	10	10	40	0.20
2000	16	26	24	0.40
3000	13	39	11	0.54
4000	8	47	3	0.73
5000	3	50	0	1.00



Hazard Rate Function

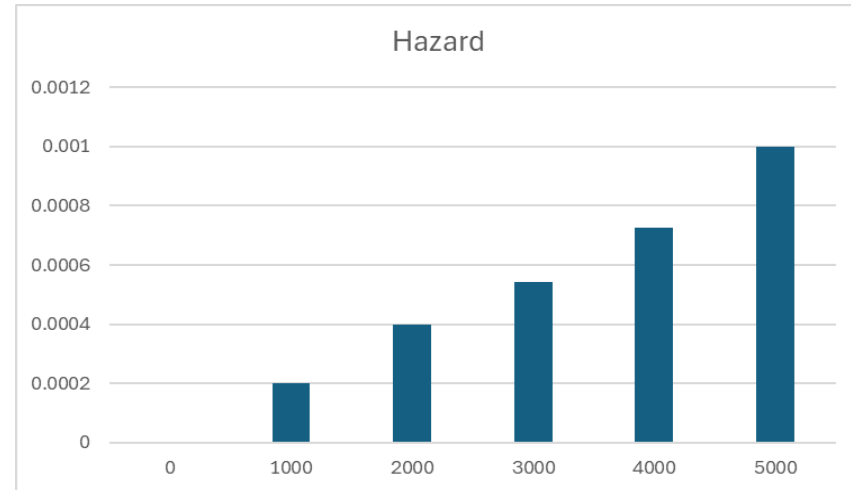
The failure rate is more typically considered as the **hazard function**, expressed as a rate.

$$h_i = \frac{pdf_i}{R_{i-1}}$$

It is the same as the failure rate, but divided by τ .

$$h_i = r_i / \tau.$$

	<i>PDF</i>	<i>CDF</i>	<i>R</i>	<i>h</i>
Hours	PDF	CDF	Rel	Hazard
0	0	0.00	1.00	0
1000	0.00020	0.20	0.80	0.00020
2000	0.00032	0.52	0.48	0.00040
3000	0.00026	0.78	0.22	0.00054
4000	0.00016	0.94	0.06	0.00073
5000	0.00006	1.00	0.00	0.00100





Failure and Hazard Rate

The ***failure rate*** is the fraction of survivors that will fail in a specific time period.

It is the fraction that fail, given that they have lived up to that point.

Also expressed as the ***hazard function***.

A decreasing failure / hazard rate means early life failures

A constant failure / hazard rate means random failures

An increasing failure / hazard rate means end of life wear out

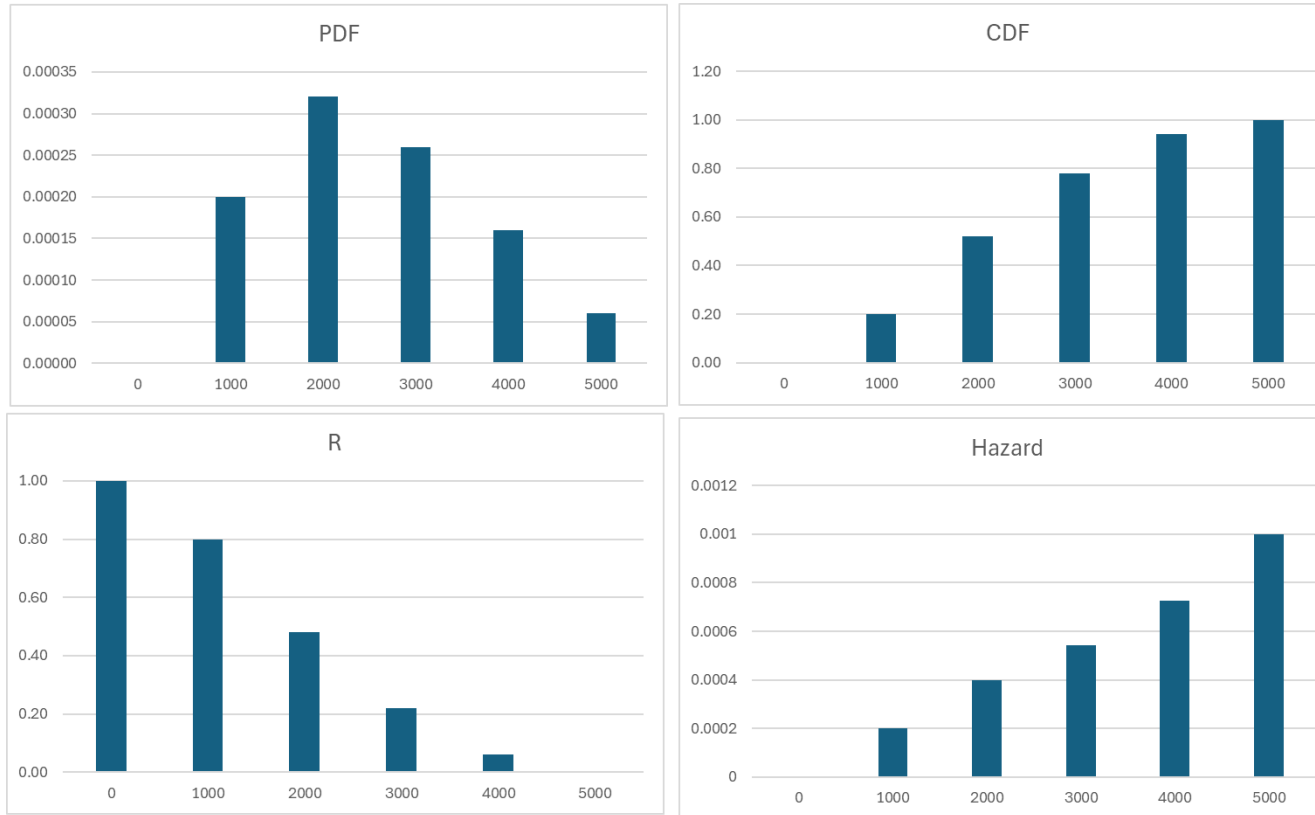
The Data as Counts

The count histograms and graphs can show the information.



Reliability Functions

These normalized histogram graphs of the 4 reliability functions are example results of a ***non-parametric reliability analysis***. Distribution free.





Key Reliability Metrics – Mathematics

Probability density function (failures)

Cumulative distribution function (failed)

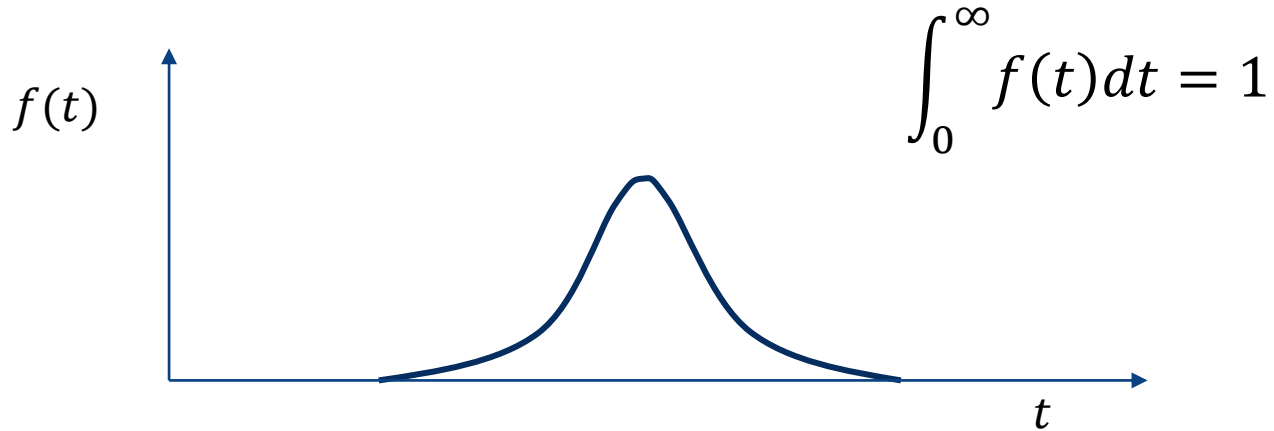
Reliability function (survived)

Hazard function (failure rate)

Instead of a **distribution free** histogram-based analysis, we might instead choose a function to represent any of these, for example a normal distribution function. Such function would include **parameters**. You would then be doing a distribution-based **parametric analysis**.

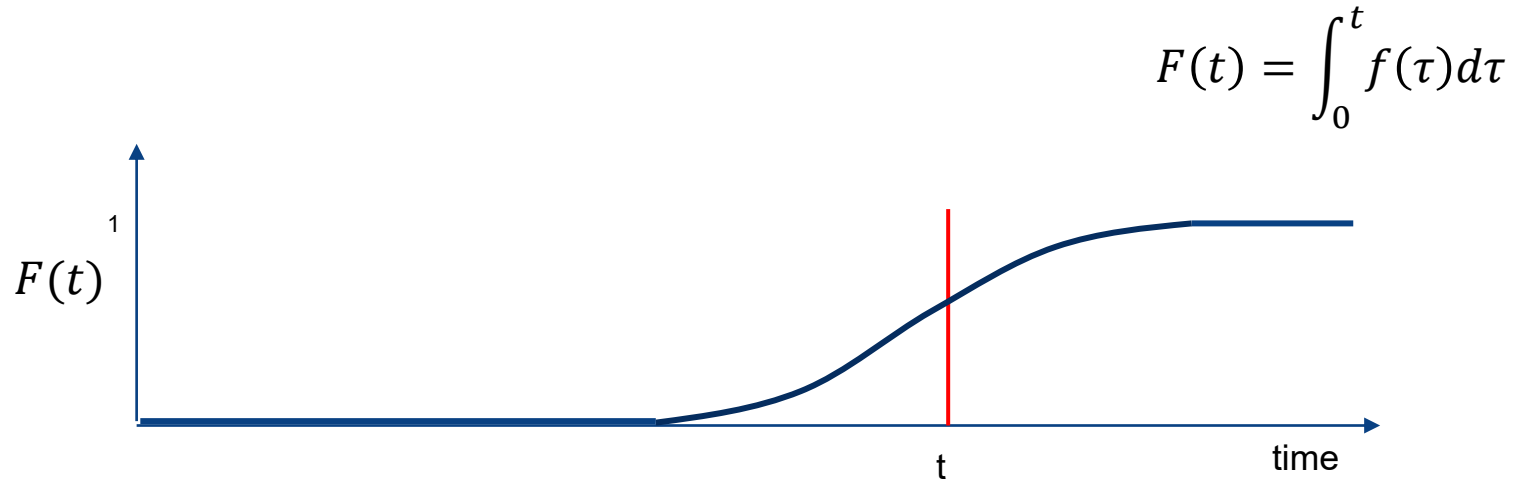
Probability Density Function

The ***probability density function*** $f(t)$ provides the distribution of life for a component. The total area under the pdf is one.



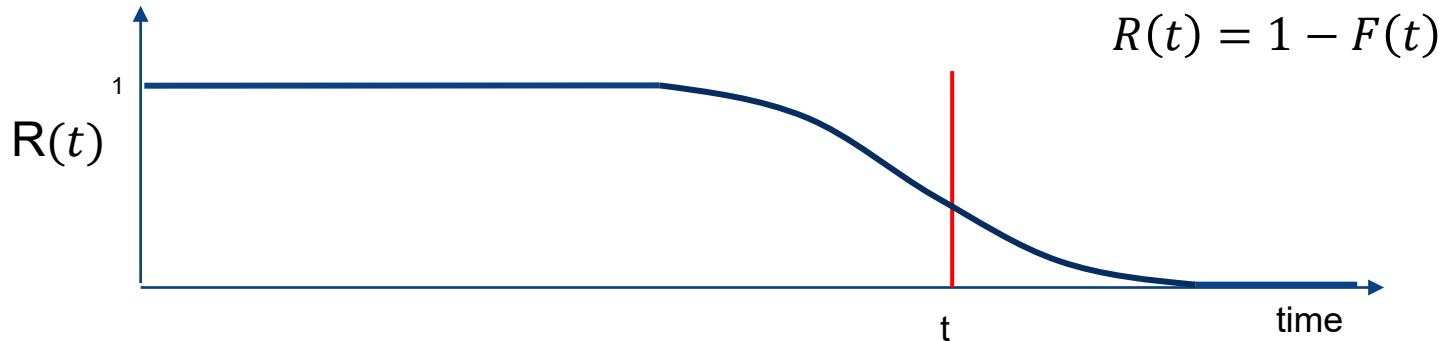
Cumulative Distribution Function

The ***cumulative density function*** $F(t)$ is the probability that a component will fail by time t . Indicates the fraction that have failed at time t .



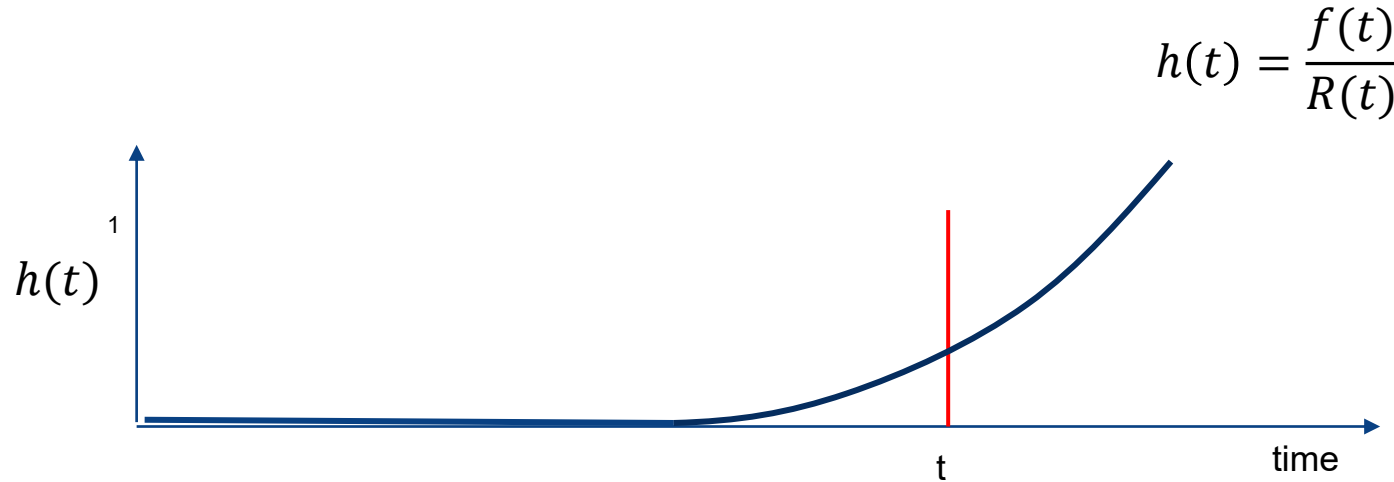
Reliability Function

The **reliability function** $R(t)$ gives the fraction surviving at any time and expresses the probability that a component will survive until time t . Indicates the survival fraction.



Reliability Function

The ***hazard function*** $h(t)$ gives the fraction dying at any time of those that have survived to that point. Indicates the instantaneous rate of failing.





Mathematical Relationships

The following relationships can be derived amongst the reliability functions.

$$F(t) = \int f(t)dt \quad f(t) = -\frac{dR(t)}{dt} \quad f(t) = h(t)e^{-\int h(t)dt}$$

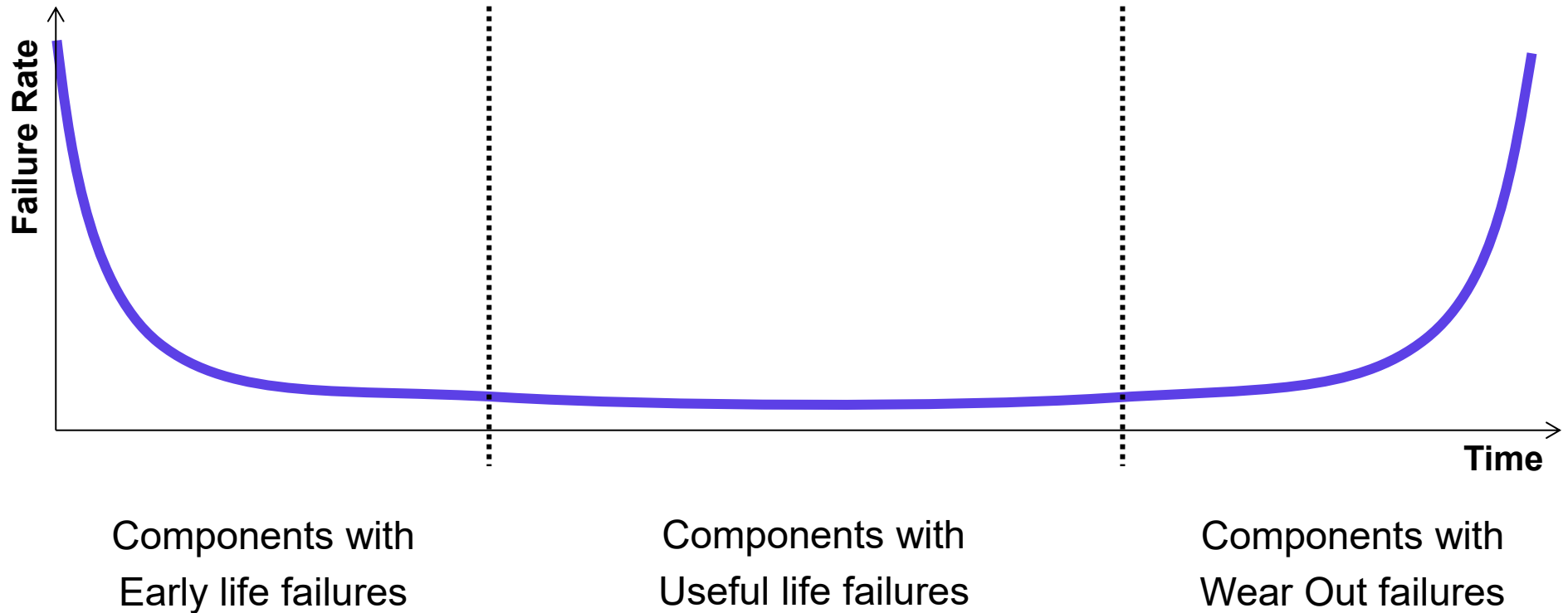
$$R(t) = 1 - F(t) \quad R(t) = e^{-\int h(t)dt} \quad \frac{dF(t)}{dt} = -\frac{dR(t)}{dt}$$

$$h(t) = \frac{f(t)}{R(t)} = \frac{f(t)}{1-\int f(t)dt} \quad -\ln[R(t)] = \int h(t)dt \quad h(t) = \frac{\frac{d}{dt}F(t)}{1-F(t)}$$

Reliability Engineering

The failure rate of a **system** generally follows a ***bath tub curve***.

This is due to the many different failure modes possible in a system.



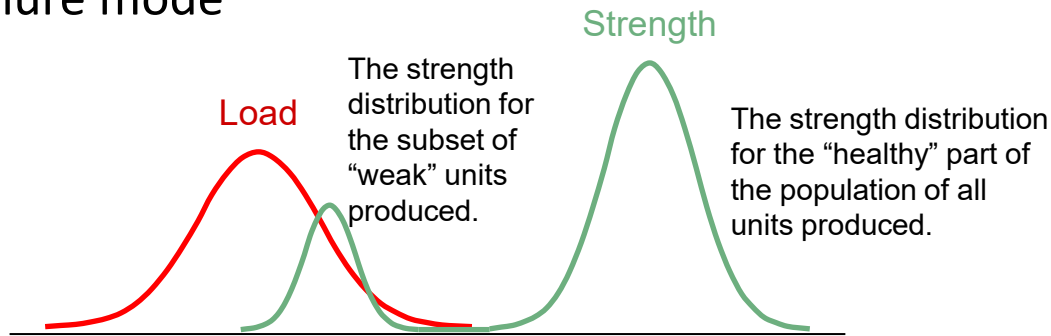
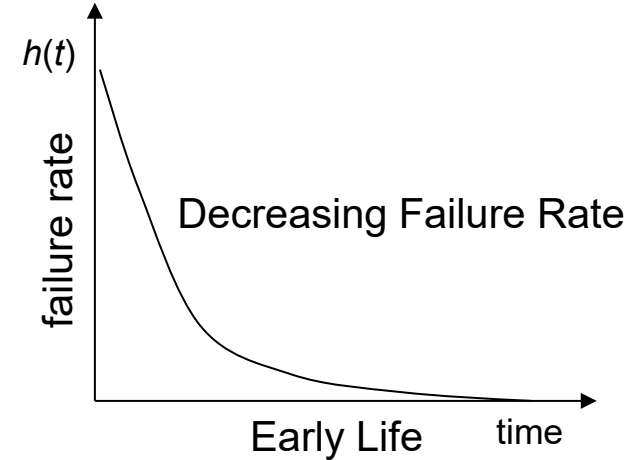
Reliability Engineering

Early life failures are due to manufacturing or material defects. Quality problems.

More likely to fail earlier, less likely to fail as time goes on.

Characterized by a decreasing hazard function

- The longer it operates, the less likely of an infant mortality failure mode



The bi-modal strength distribution for a population of devices with a weak subset.



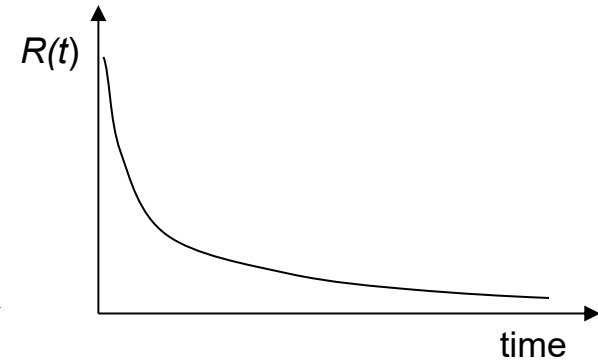
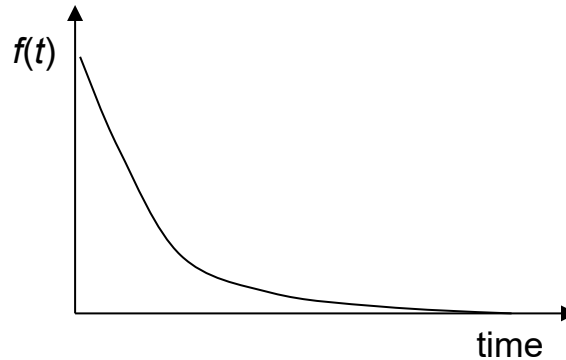
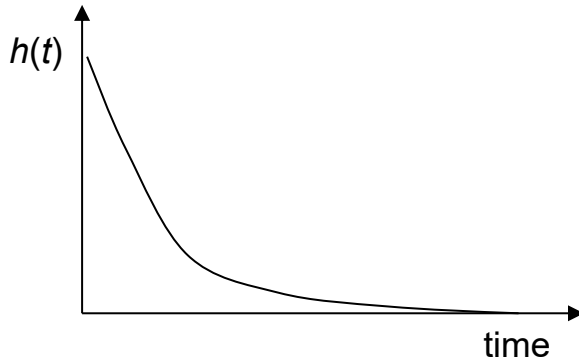
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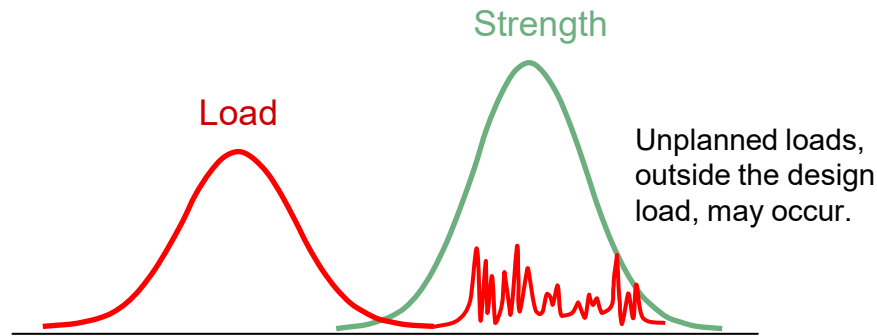
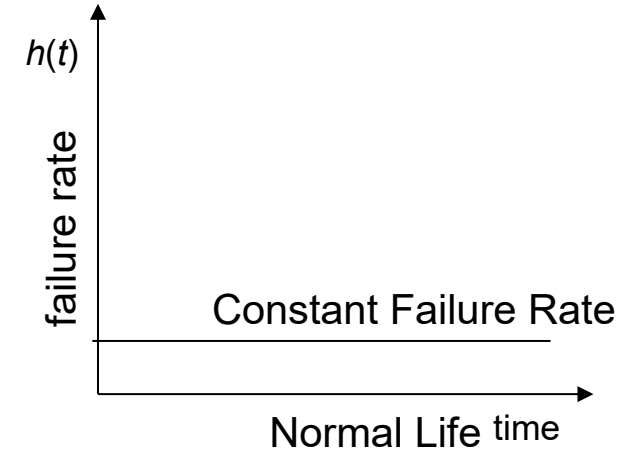
Reliability Engineering

Normal life failures due to shock load failures

Equally likely to occur at any moment

Characterized by a flat hazard function

- The failure rate is independent of time



A design load distribution, with unanticipated high loads occasionally observed



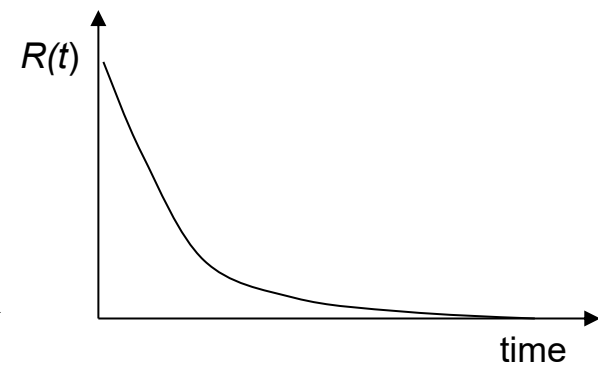
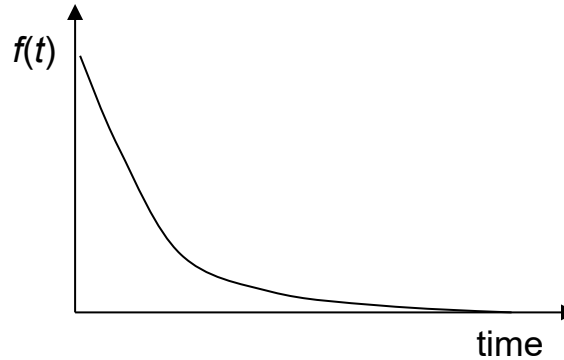
Reliability Engineering

Normal life failures due to shock load failures

Equally likely to occur at any moment

Characterized by a flat hazard function

- The failure rate is independent of time



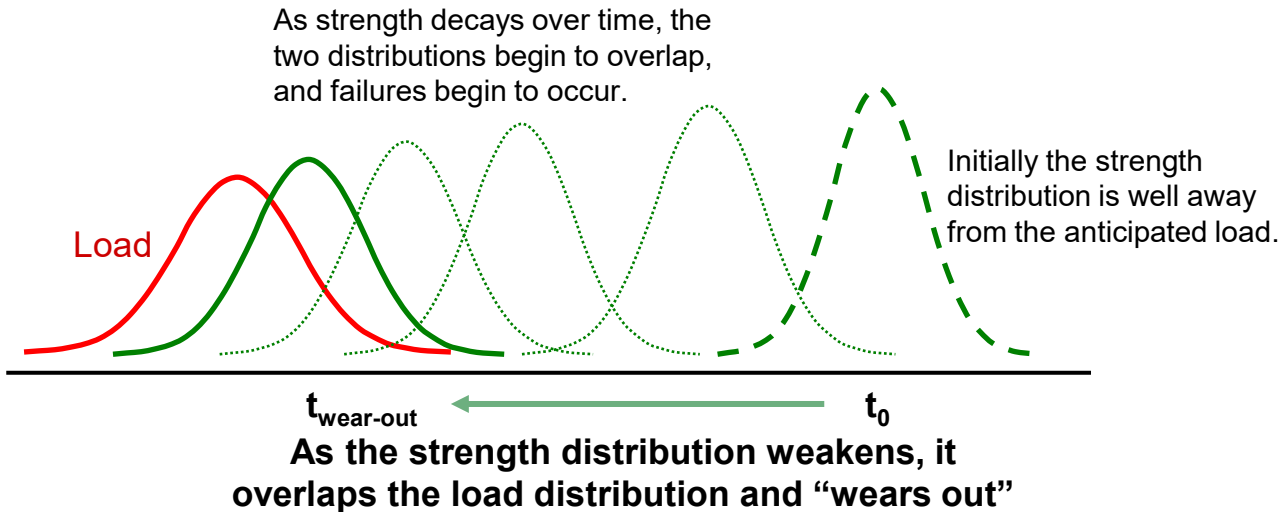
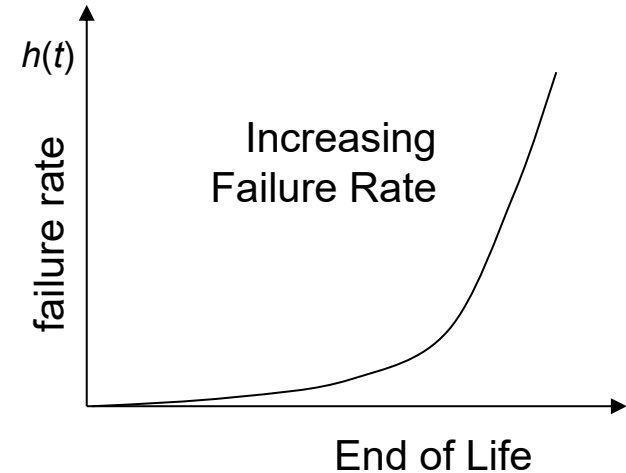
Reliability Engineering

End of life failures due to wear out or fatigue

Do not happen early in life, more likely to occur as time goes on

Characterized by an increasing hazard function

- The longer it operates, the more likely of wearing out



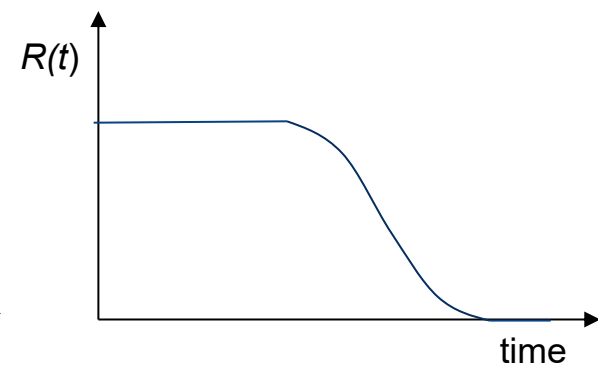
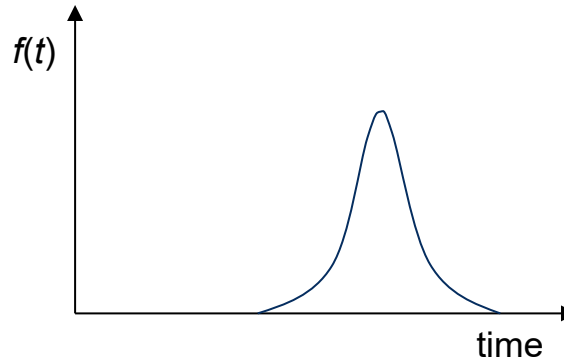
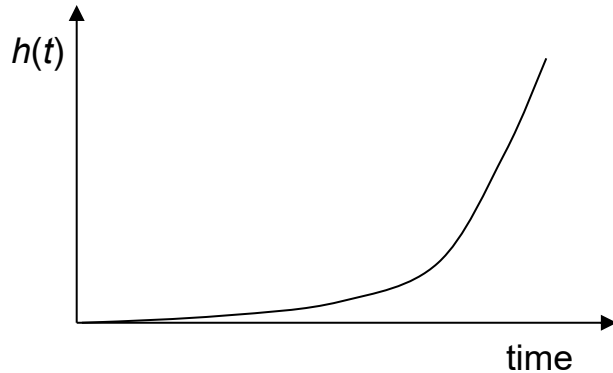
Reliability Engineering

End of life failures due to wear out or fatigue

Do not happen early in life, more likely to occur as time goes on

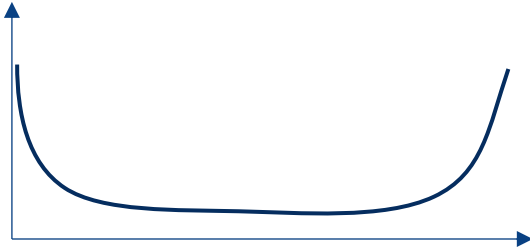
Characterized by an increasing hazard function

- The longer it operates, the more likely of wearing out



Distribution Types and Physics

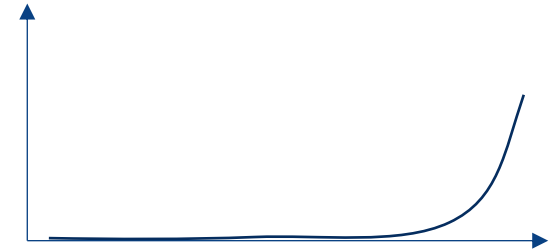
Your perceived physics of failure should align with the shape of the hazard function!



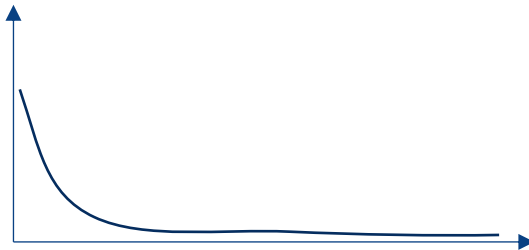
Bathtub: many mixed failure modes



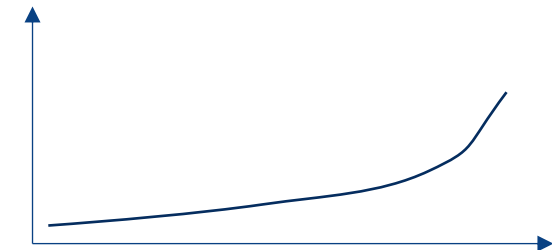
Flat: random excess loads



End of life: wear out



Early life: manufacturing defects



Increasing: possibly fatigue related



Overview

Reliability

Weibull distribution

System Reliability

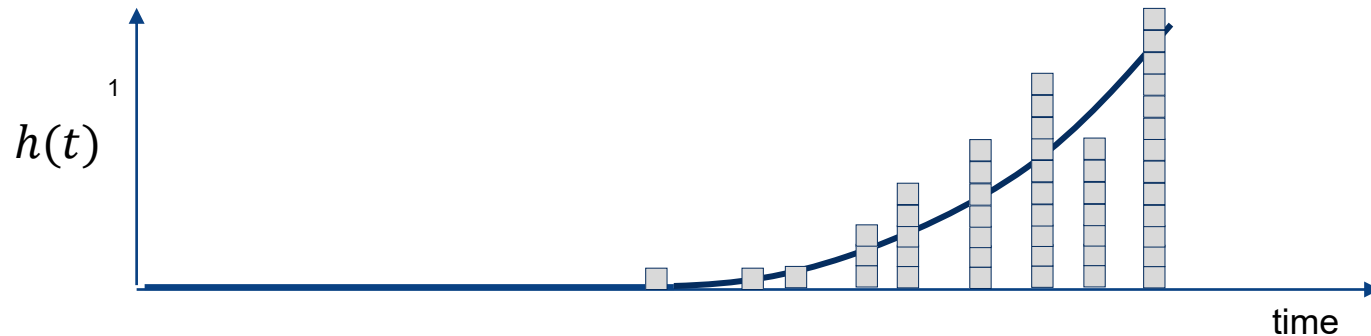
MTBF

Hazard Rate

Maintenance Plans

Distribution Fitting

Of the distributions, the hazard function is the one with data structure that “looks” like typical functions – $\exp()$, $\ln()$, etc., and also looks different for the different failure modes. So to fit a distribution we often start with this.





Weibull Distribution

A useful distribution function for reliability purposes is the **Weibull distribution**, expressed as

$$h(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1}$$

This is the two parameter form, β and η .

Notice it is essentially an exponential function of time.

β is known as the shape parameter.

η is known as the scale parameter.

Weibull Distribution

The other reliability functions become

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

$$F(t) = 1 - e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

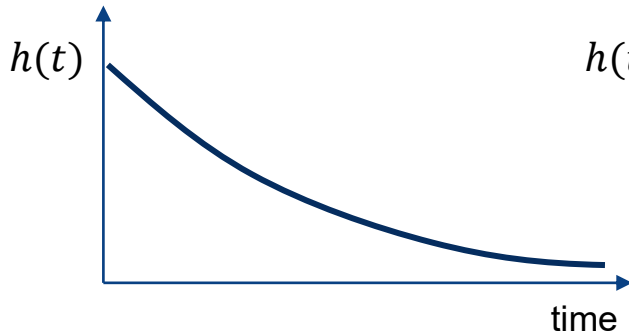
$$R(t) = e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

Weibull Distribution

With this parametric form, the distribution handily fits the different failure mode types.

$$h(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1}$$

$$\beta < 1$$



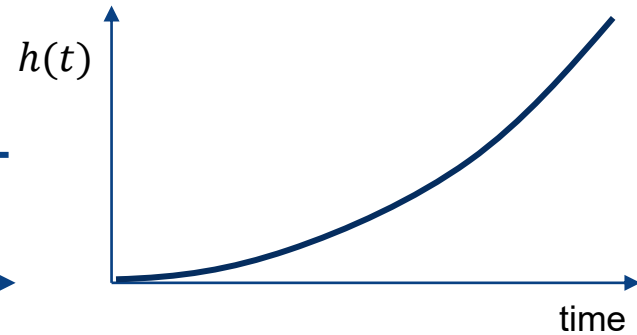
Early Life Failure

$$\beta = 1$$



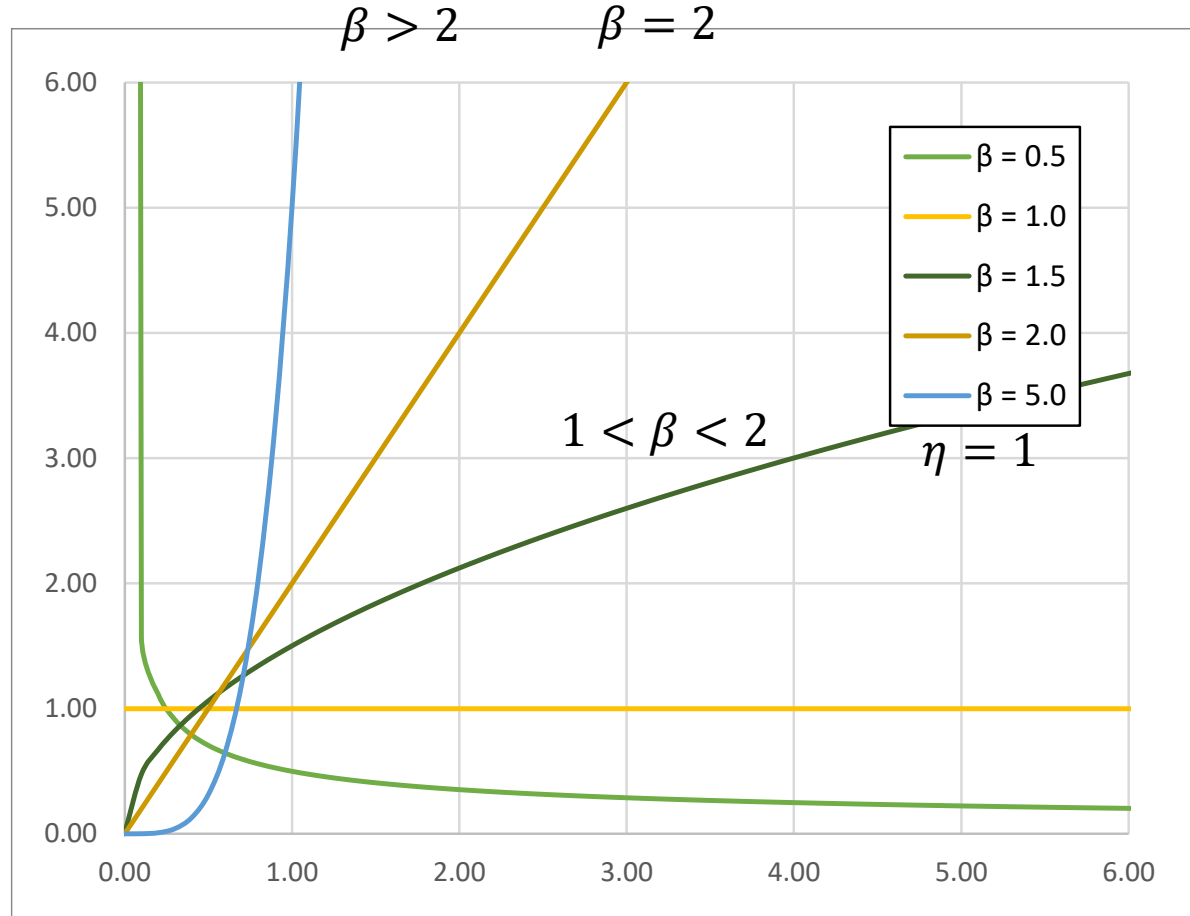
Normal Life Failure

$$\beta > 1$$



End of Life Failure

Hazard Function and Shape Parameter

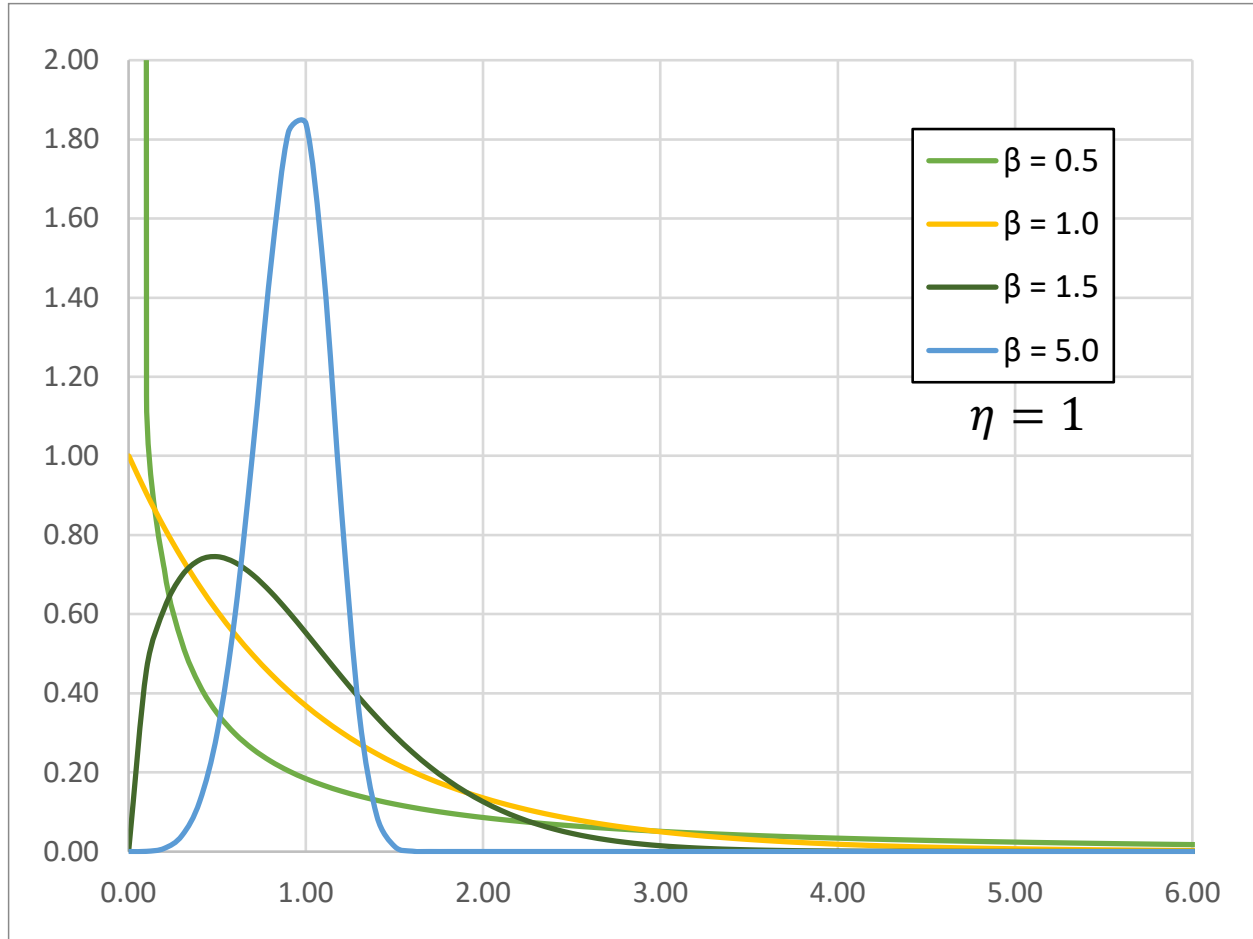


$$h(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1}$$

$\beta = 1$

$\beta < 1$

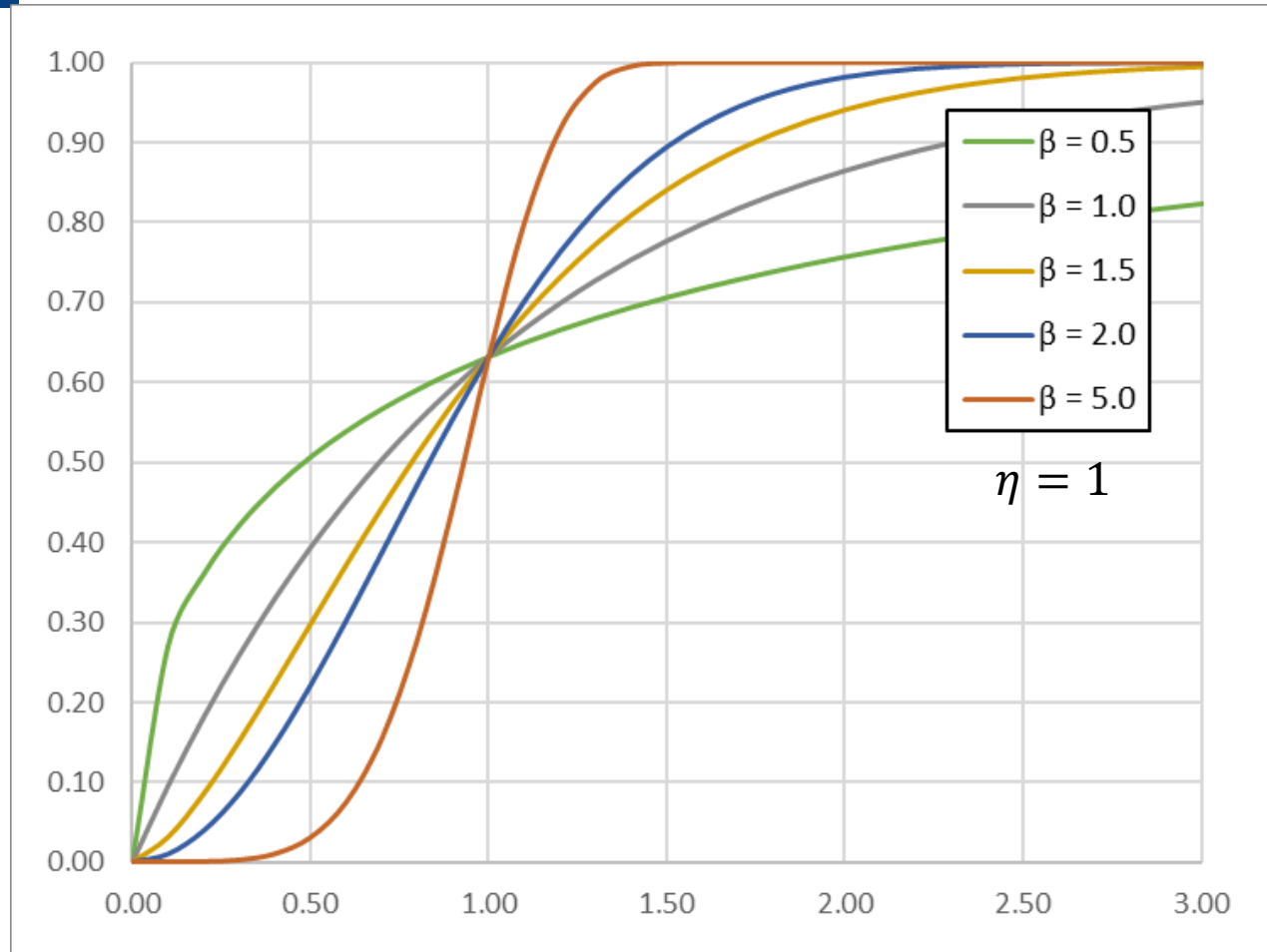
Probability Density Function and Shape Parameter



$$h(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1}$$

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

Cumulative Density Function and Shape Parameter



$$h(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1}$$

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

$$F(t) = 1 - e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$



Reliability Functions Summary

There are four useful reliability functions to express a part's reliability: failure density, cumulative failures, the hazard function and the reliability function.

Among the four functions, the hazard function is most easy to associate to the three types of failure modes and often the starting point for assuming a relationship to failure data.

The Weibull distribution can mathematically represent most types of failure modes, whether early life, normal life or end of life.



Overview

Reliability

Weibull distribution

System Reliability

MTBF

Hazard Rate

Maintenance Plans

What does reliability look like for a system?

Suppose you have a complex electro-mechanical system or process.

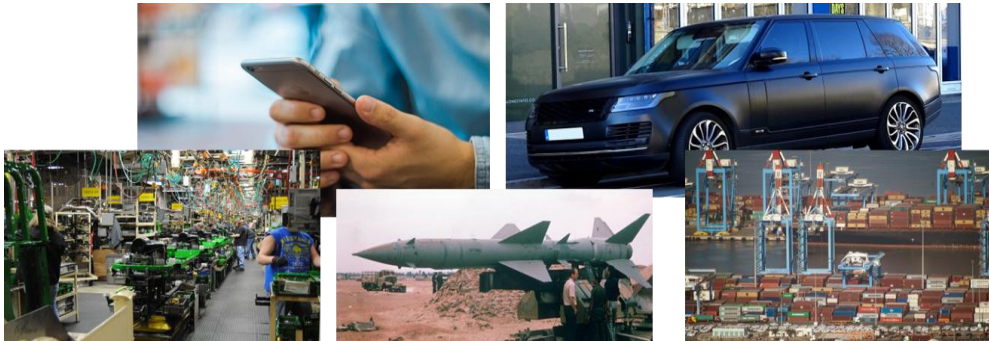
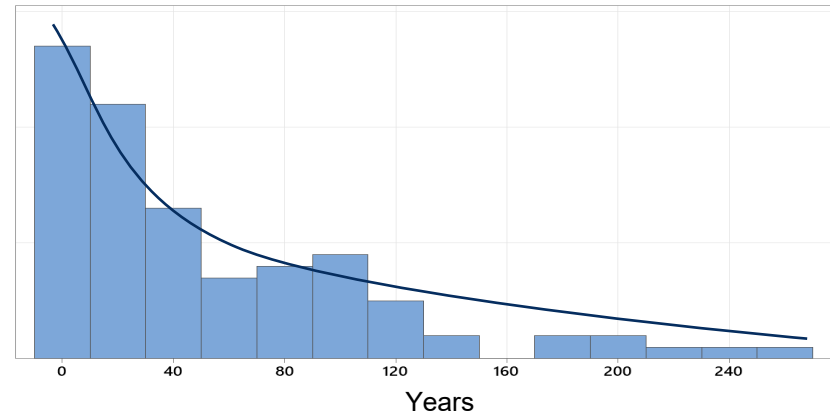
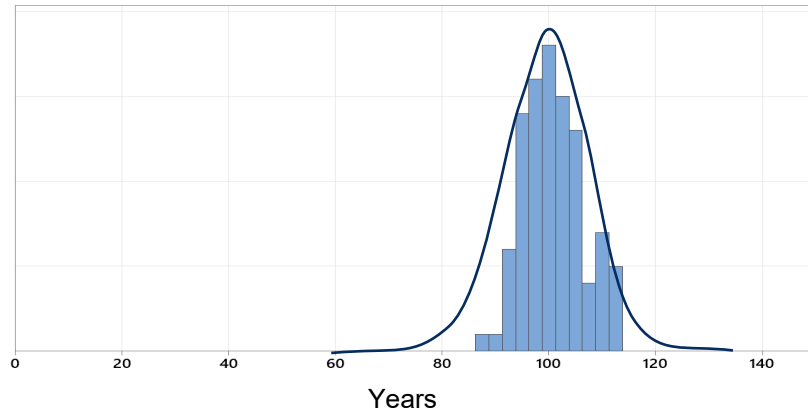
What does a system reliability life distribution function (PDF) look like?



What does reliability look like for a system?

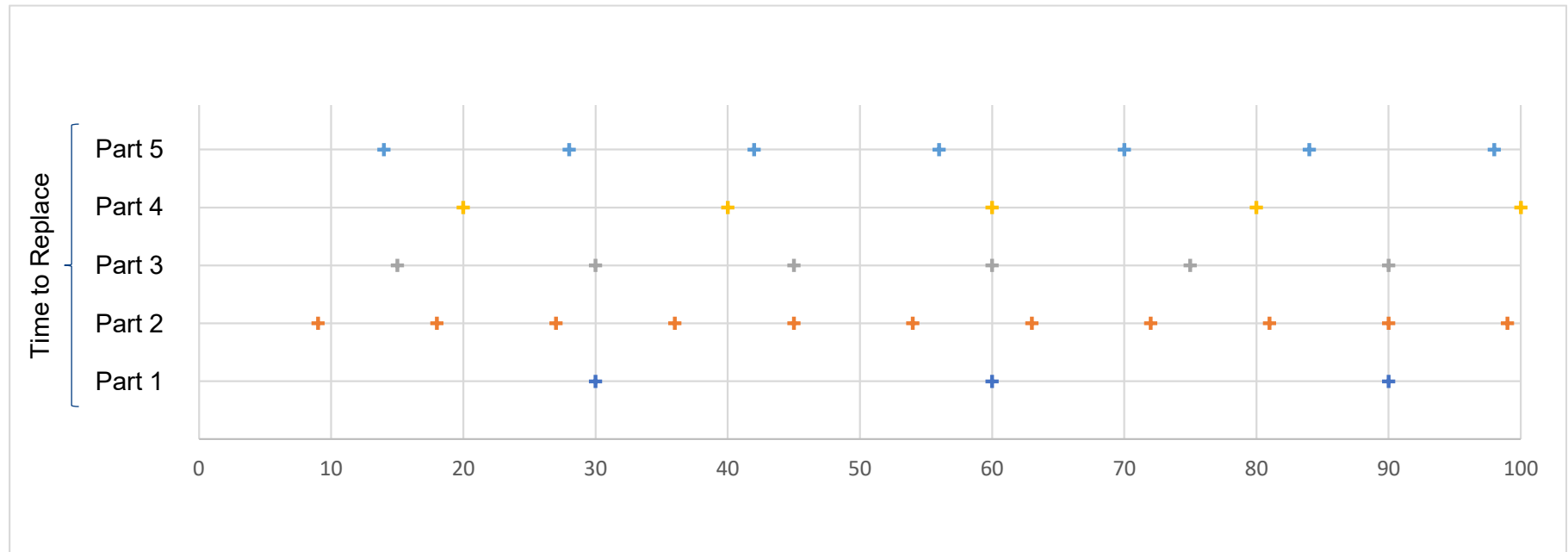
Suppose you have a complex electro-mechanical system or process.

Which of the two does a system reliability life distribution function (PDF) look like?



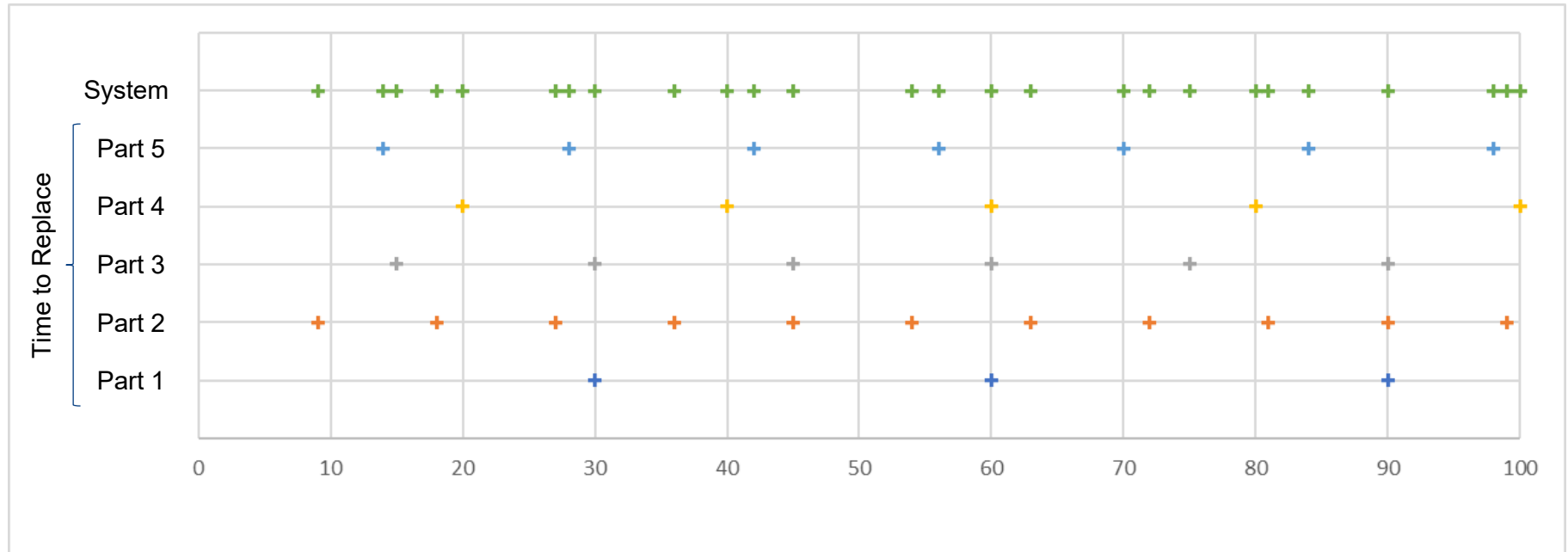
What does reliability look like for a system?

Suppose you have a complex electro-mechanical system. 5 parts, each with different life. What does a system reliability life distribution function (PDF) look like?



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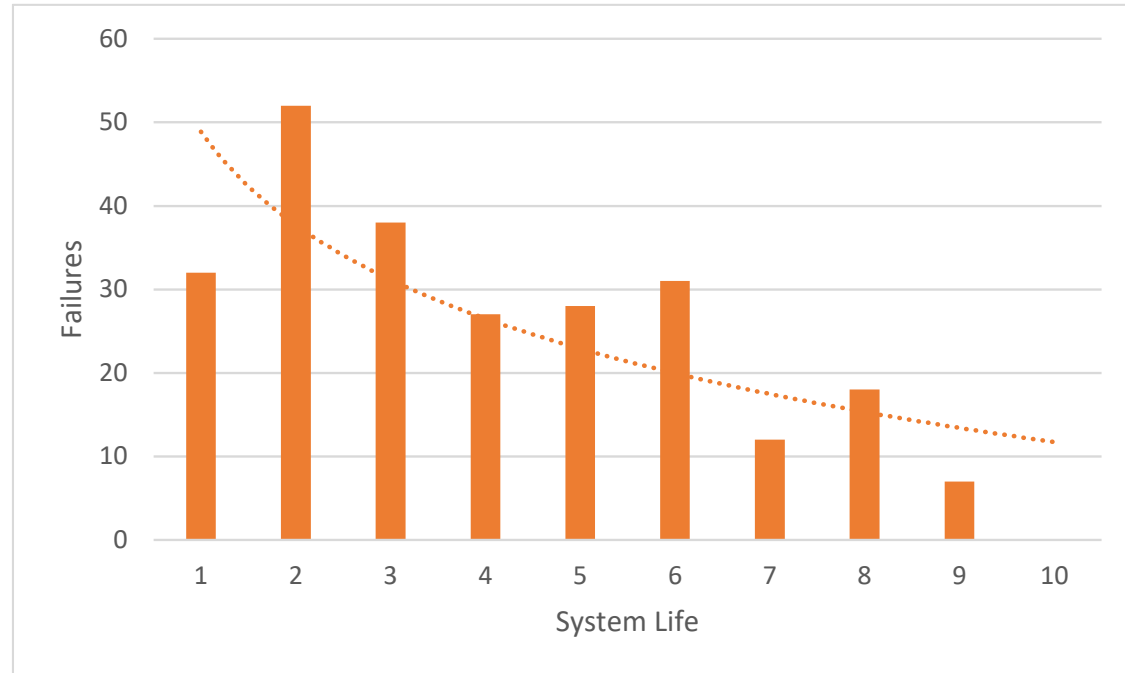


What does reliability look like for a system?

Suppose you have a complex electro-mechanical system.

What does a system reliability life distribution function (PDF) look like?

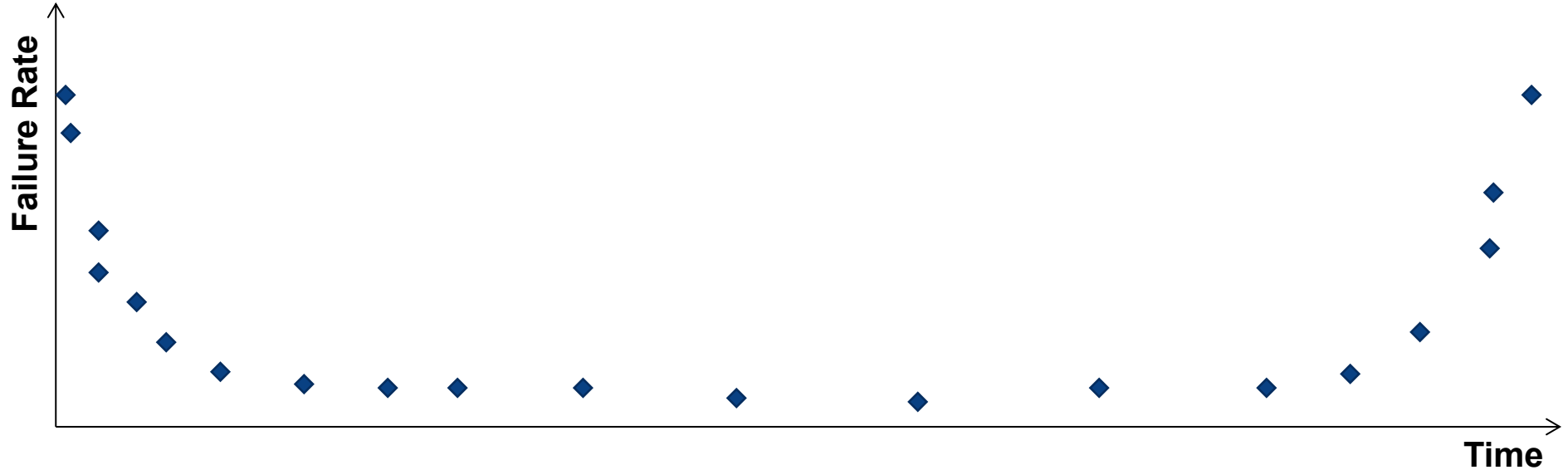
All systems tend toward an exponential life distribution as the number of parts increases.



What does reliability look like for a system?

A system of many parts exhibits a bath tub curve for the hazard function.

What is the best fit Weibull function to a *bath tub curve* shaped hazard dataset?

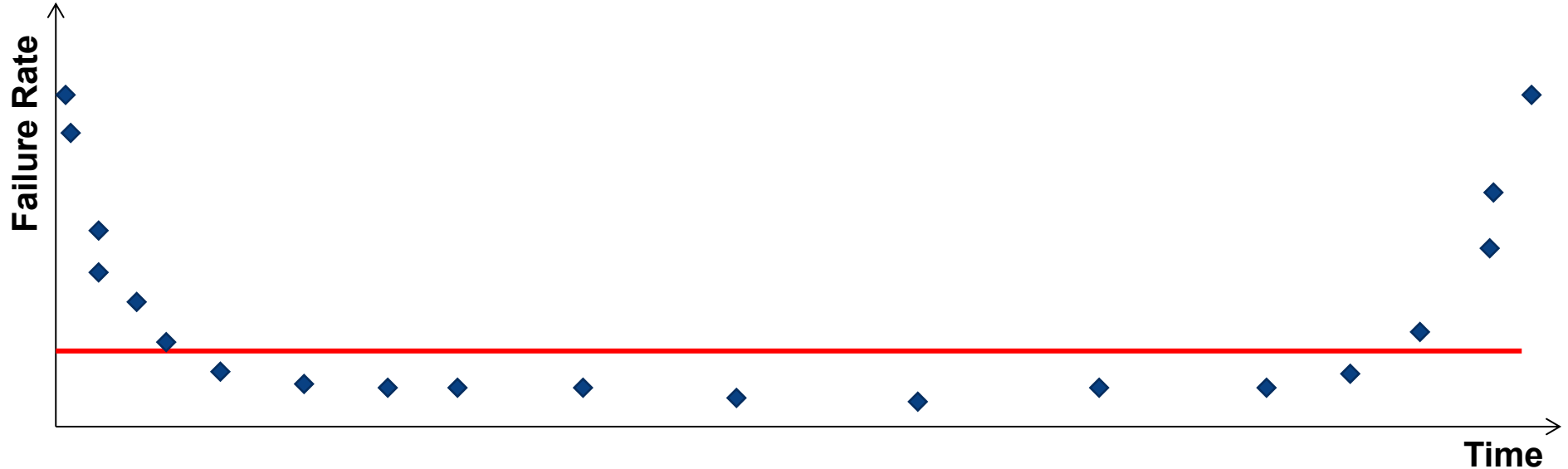


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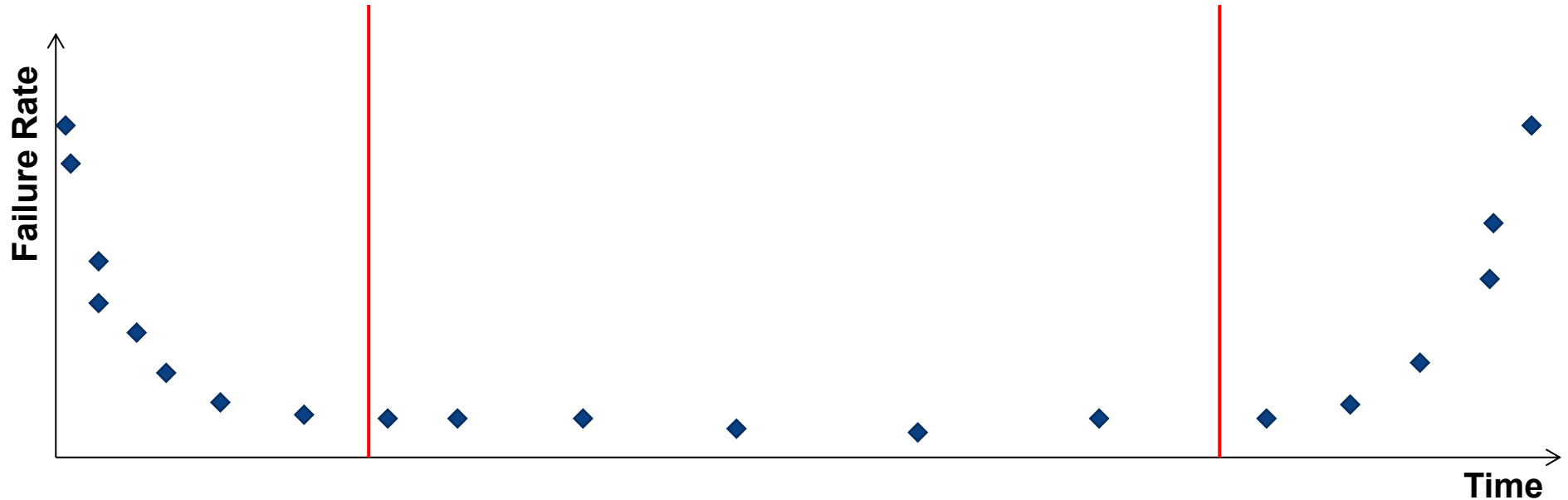
What is the best fit Weibull function to a *bath tub curve* shaped hazard dataset?

A flat line. A flat hazard function corresponds to an exponential life function (PDF).



What does reliability look like for a system?

If you see multiple modes in your life hazard data, split it apart and analyze separately.



System Reliability Distributions

Given a system has an exponential hazard function, system reliability functions can often fit and be considered as an ***exponential distribution***.

$$h(t) = \lambda$$

$$f(t) = \lambda e^{-\lambda t}$$

$$F(t) = 1 - e^{-\lambda t}$$

$$R(t) = e^{-\lambda t}$$

The only parameter is λ , the **average failure rate**, or $1 / (\text{mean time to failure})$. Then,

$$\lambda = -1/MTTF$$

$$MTTF = -1/\lambda$$

Exercise – Space System

Suppose a Mars rover has a mission life of 90 days.

Suppose an acceptable risk level is 2% chance of failure during mission life.

Given the system failure rate is exponential, what is the rover's life distribution function equation?

What is the mean time to failure of the rover?

What is the ratio of mission life versus mean time to failure?



Living Beyond the Extended Warranty

With a mission planned for 90 sols (Martian days), the Mars Exploration Rovers Spirit and Opportunity lived way past their "warranty dates." Spirit lasted thousands of days longer than the original plan. Opportunity has lasted 5,000+ days on Mars! So, how have these rovers survived the rocky terrain, the frigid cold, and the wind-blown dust scouring the surface of Mars?

Exercise – System

Consider the following failure rate data of a production line, shown are days until a failure occurred and a shutdown was required.

Complete a distribution-free analysis and plot the hazard rate.

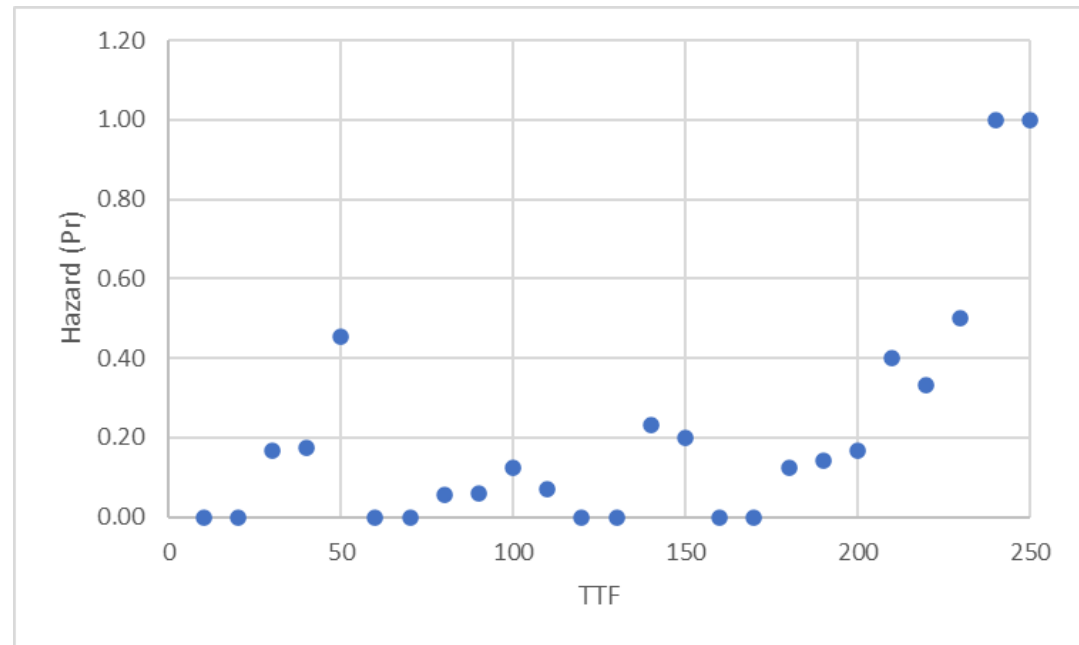
Comment on its shape.

46	108	182	24
229	29	45	41
45	207	150	220
32	145	140	41
30	22	35	47
238	135	131	46
210	31	35	44
46	25	83	192
21	99	78	27
95	37	44	41
21	44	43	43
39	37	45	175

Exercise – System

Hrs	Failed	Failures	Alive	Hazard
10	0	0	48	0.00
20	0	0	48	0.00
30	8	8	48	0.17
40	15	7	40	0.18
50	30	15	33	0.45
60	30	0	18	0.00
70	30	0	18	0.00
80	31	1	18	0.06
90	32	1	17	0.06
100	34	2	16	0.13
110	35	1	14	0.07
120	35	0	13	0.00
130	35	0	13	0.00
140	38	3	13	0.23
150	40	2	10	0.20
160	40	0	8	0.00
170	40	0	8	0.00
180	41	1	8	0.13
190	42	1	7	0.14
200	43	1	6	0.17
210	45	2	5	0.40
220	46	1	3	0.33
230	47	1	2	0.50
240	48	1	1	1.00
250	48	0	0	1.00

This is system level failure data, likely with multiple failure mode sources. Looking at the data, it seems there are 2 failure modes, one early, 0-50 days, and then one longer, 100-200 days.

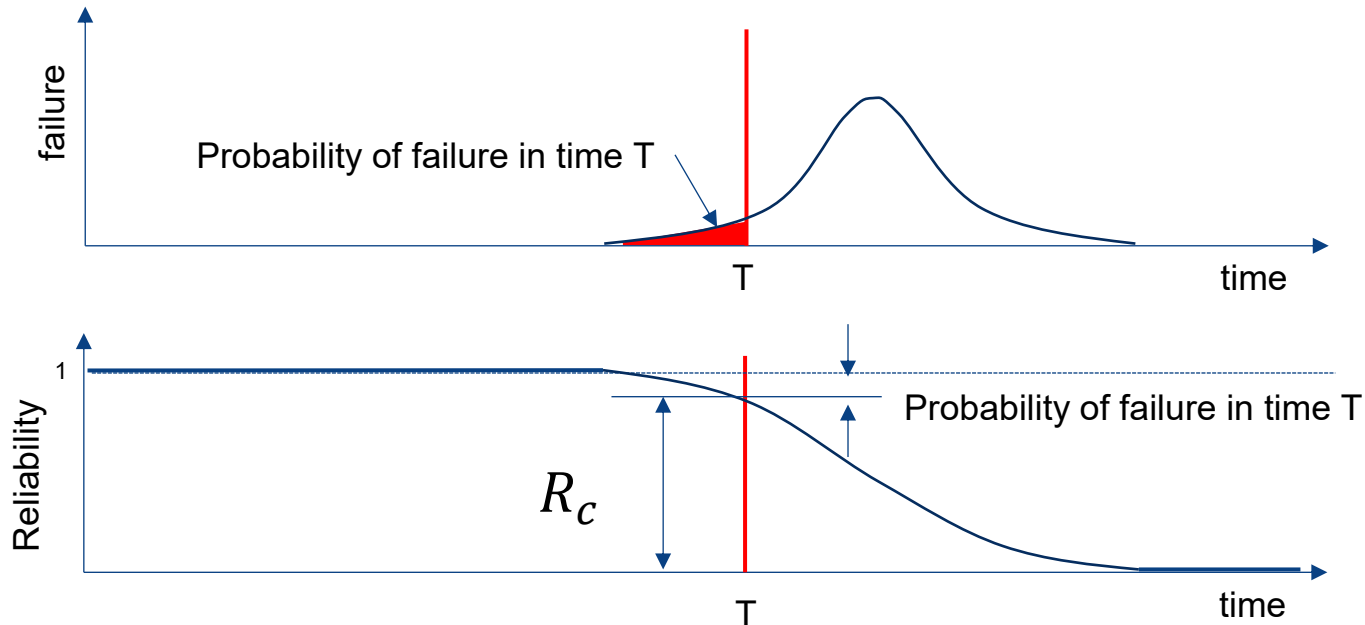


Reliability

Probability-of-failure and reliability of a component are distribution functions of time.

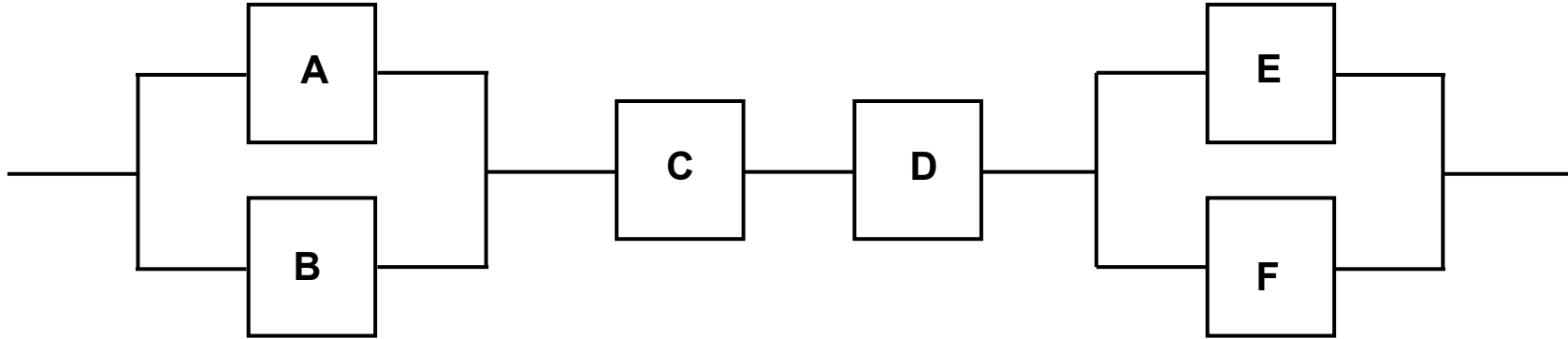
If we define a time period T then we can define the **reliability** R_c of a component.

We will now use these component reliabilities R_c in considering systems.



System Reliability Block Diagrams

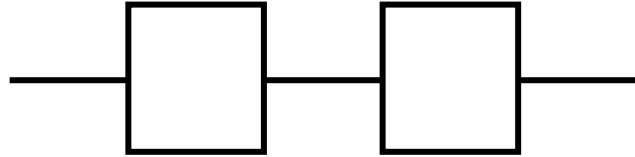
We can model a system with a simple reliability block diagram



Reliability block diagrams are useful as models to understand system reliability R_s in terms of components and their component reliabilities R_i .

Reliability Block Diagrams

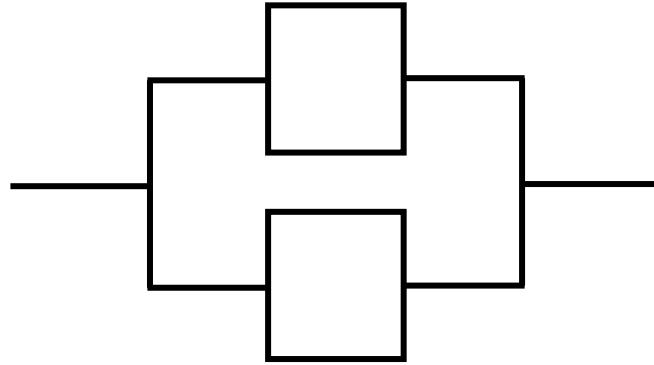
Blocks in series represent the scenario when the system fails if either element fails.



Reliability Block Diagrams

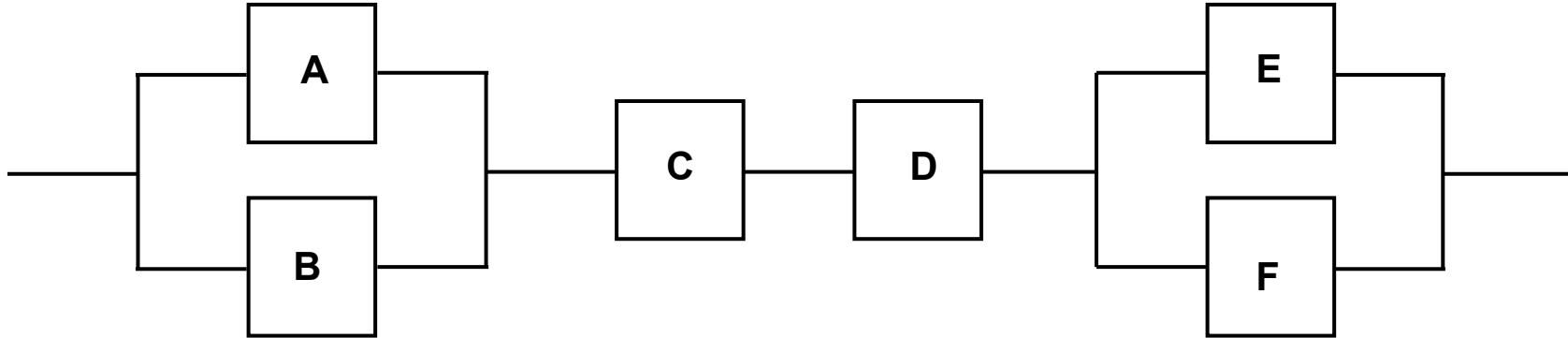
Two blocks in parallel represent when both elements must fail for the system to fail.

You can use either. There is *redundancy*.



Reliability Block Diagrams

We can model a system with a simple reliability block diagram



For a system to fail: (A and B) or (C) or (D) or (E and F) must fail.

Example – Chiller Plant

A chiller plant provides chilled water to a facility.

This plant has 2 compressors, 2 valves, 2 evaporators 5 condenser sections, and 1 controller.

What is the plant reliability for part load operation, when only one of each is needed?

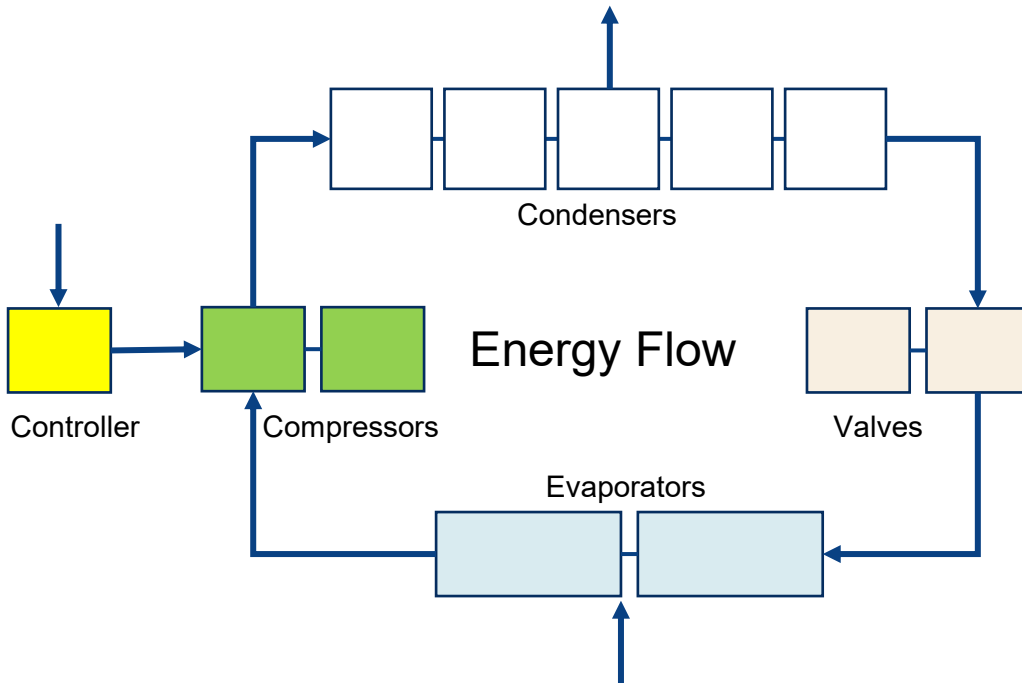


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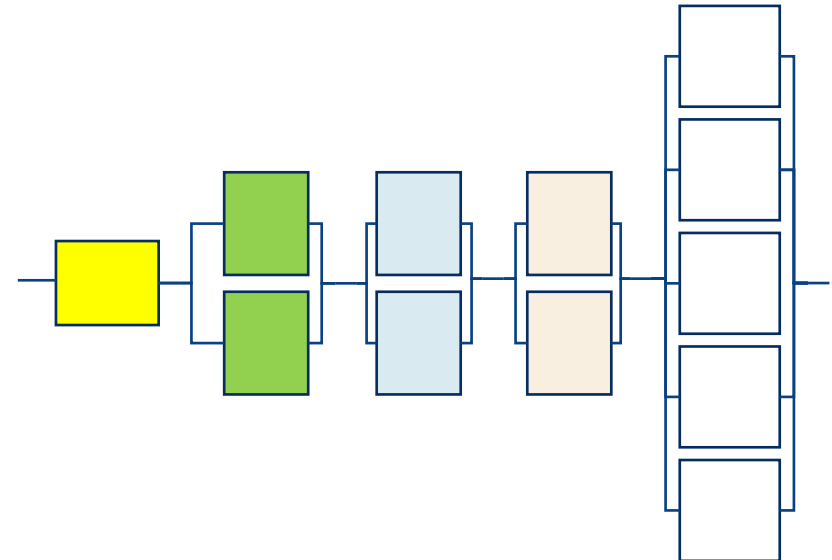
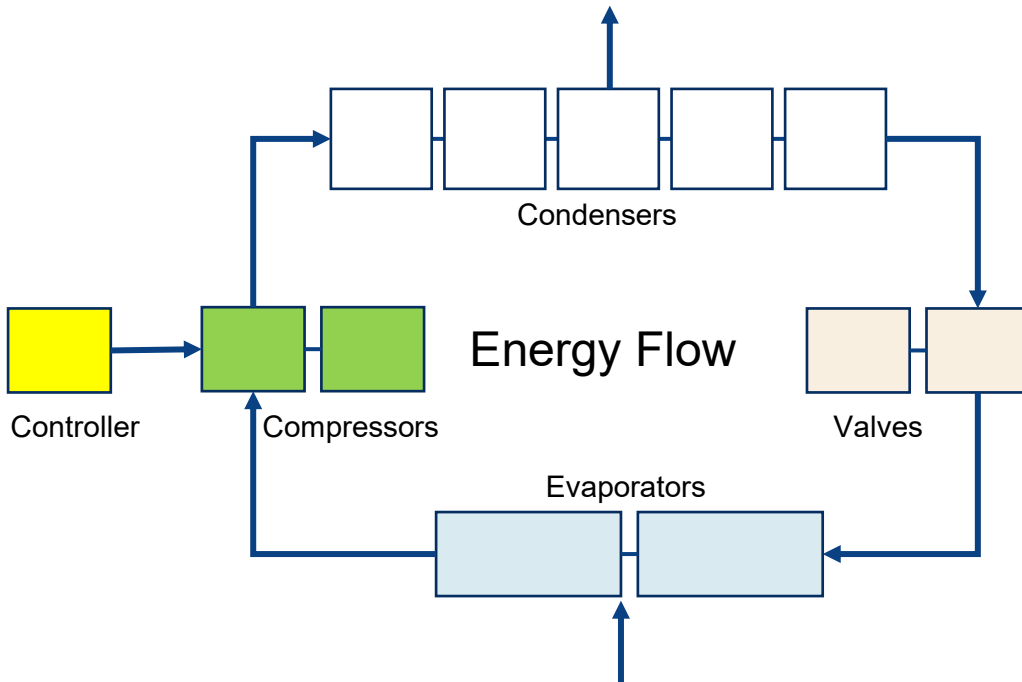


Example – Chiller Plant

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Part Load Reliability



Availability

When we have redundant systems, operating with a failed redundant part allows the system to remain operational. The system remains available.

The ***system availability*** A_s is the reliability of a system using any redundant option.

Often, an available system with a failed redundant part only operates a partial capacity.



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Often, an available system with a failed redundant part only operates a partial capacity.

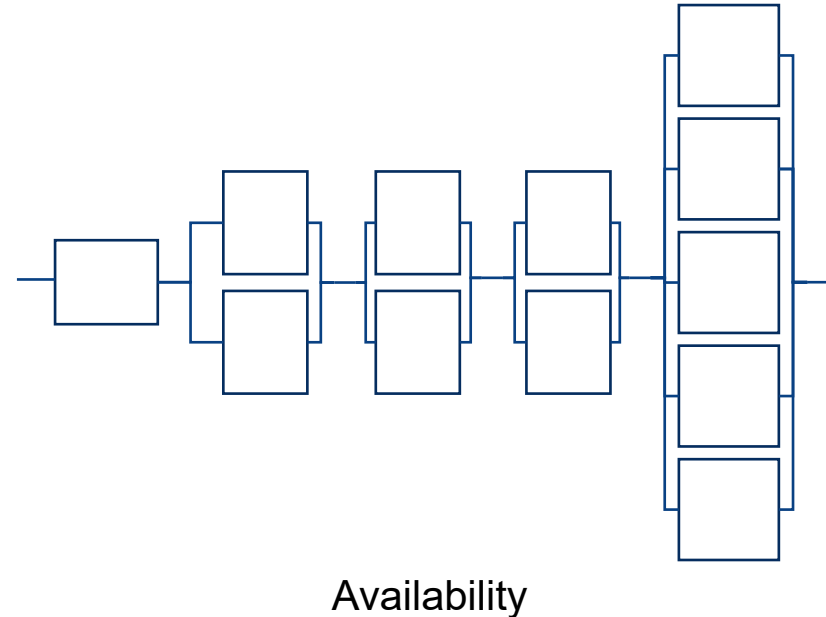
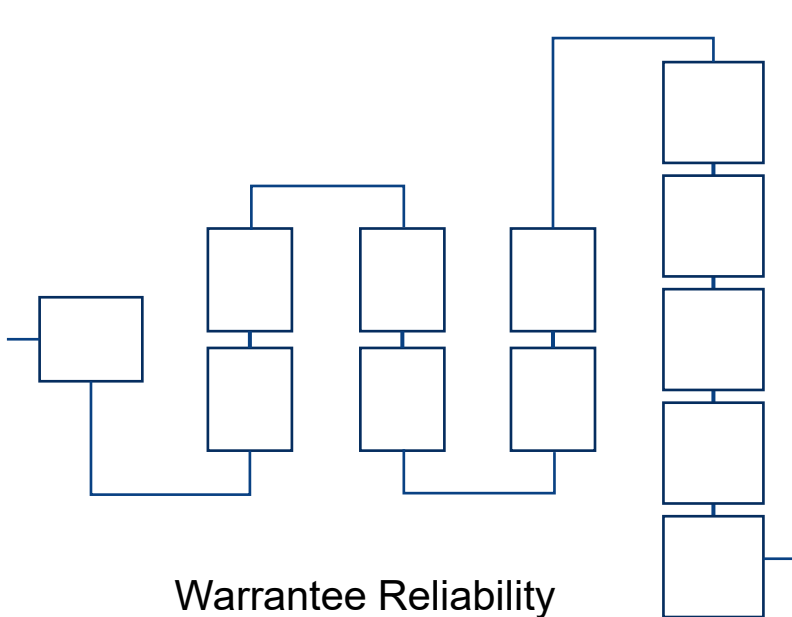
The customer will not be satisfied with a failed redundant part, it must still be fixed.

Therefore we often call the *system reliability* R_s as the *series evaluated reliability* or **warranty reliability** of all components, representing the reliability with no servicing.

Warranty

A warranty is typically offered on a system to warrant that the system will be defect free for a period of time, the warranty period. This is usually short – 3 months, 6 months, ...

To analyze the reliability in the warranty period, the system is analyzed as a series system, since any failure must be fixed.





Steps In Building Reliability Model

Given a system, how to determine the system reliability and availability?

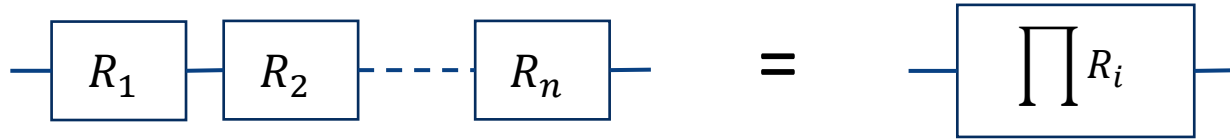
1. Compile available component reliability data
2. Build the Reliability Block Diagram
3. Form the system reliability equation or truth table

System Series Reliability

Consider a system where if any one element fails, the system fails.

The series of elements can be reduced to single equivalent system element.

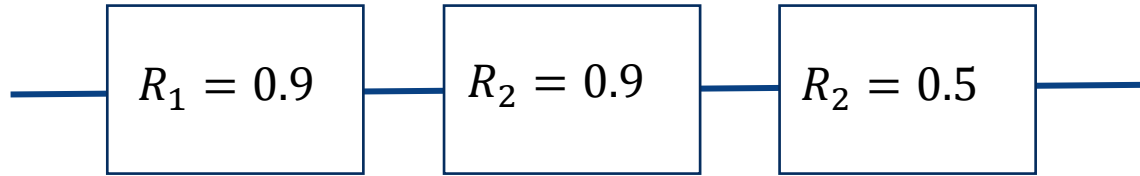
Following the probability math for independent events, the system reliability equation becomes:



$$R_s = R_1 R_2 R_3 \dots R_n$$

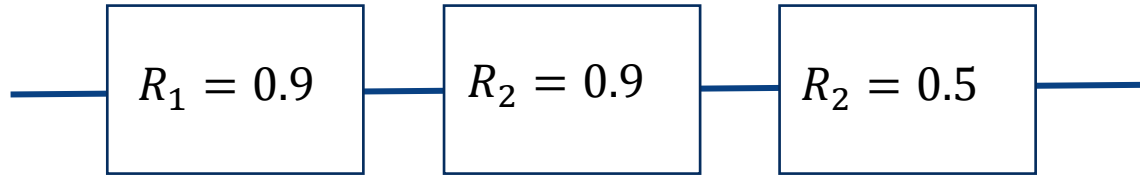
System Series Reliability Example

What's the reliability of the series system shown below?



System Series Reliability Example

What's the reliability of the series system shown below?



$$R_s = R_1 R_2 R_3 = 0.9 \cdot 0.9 \cdot 0.5 = 0.4$$

A series system's reliability is always lower than that of the lowest reliability element.

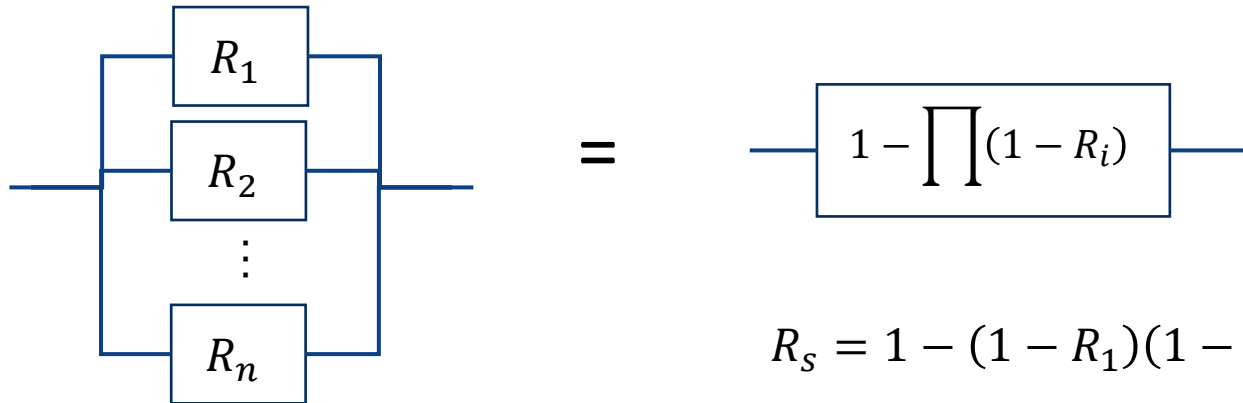
Lusser's Law

System Parallel Reliability

Consider a system where one or more elements can fail and the system will continue to function as long as one element remains operational. A *redundant* system.

The parallel set of elements can be reduced to single equivalent system element.

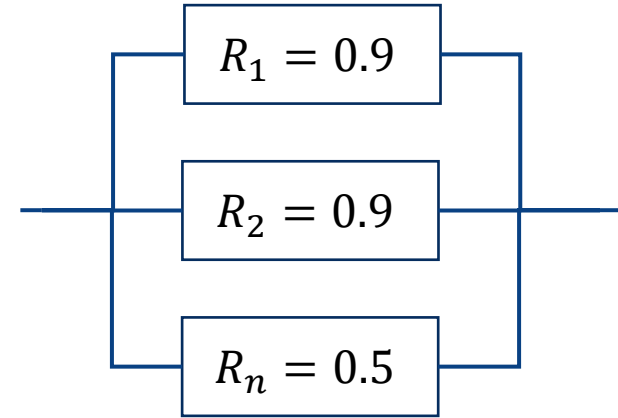
Following the probability math for independent events, the system reliability equation becomes:



$$R_s = 1 - (1 - R_1)(1 - R_2) \dots (1 - R_n)$$

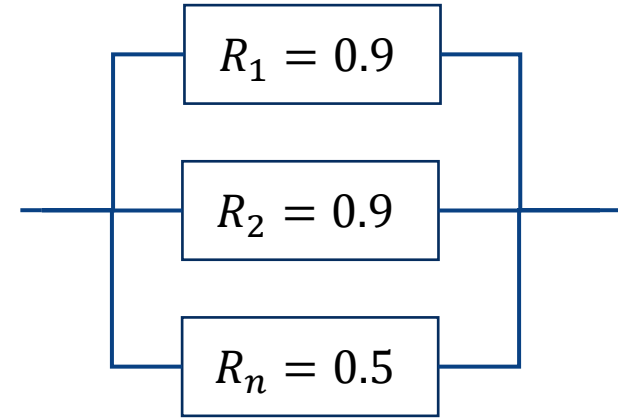
System Parallel Reliability – Example

What's the reliability of the parallel system shown?



System Parallel Reliability – Example

What's the reliability of the parallel system shown?



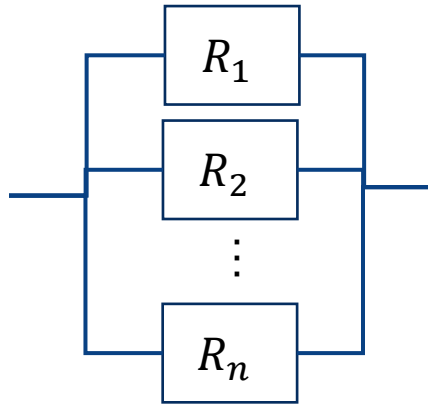
$$R_s = 1 - (1 - R_1)(1 - R_2)(1 - R_3) = 1 - (1 - 0.9)(1 - 0.9)(1 - 0.5) = 0.995$$

A redundant (parallel) system's reliability is higher than that of the highest element.

Reliability and Availability

With redundant components, the parallel computed reliability is often called the **system availability** A_s indicating the reliability of the operation of the system.

If a parallel component fails, the system doesn't fail but the failed part nonetheless needs repair. Again the consideration of warrantee or repair reliability R_s is computed as a serial system.



$$A_s = 1 - (1 - R_1)(1 - R_2) \dots (1 - R_n) \quad \text{Availability Reliability}$$

$$R_s = R_1 R_2 R_3 \dots R_n \quad \text{Repair Reliability}$$

System M of N Availability

Suppose your system must have M components working with high availability A_s . How many components N do you need if each has reliability R_c ?

That is, what is the system availability A_s defined by M of N working.

With a little math,

$$A_s = \sum_{i=1}^M \frac{N!}{i! (N-i)!} R_c^{N-i} (1 - R_c)^i$$

which is simply the cumulative binomial distribution:

$$A_s = \text{binom}(N, M, R_c)$$

System M of N Availability – Example



Suppose your system has four redundant critical components that each operate at 99.9% reliability. Suppose system failure occurs when more than two components fail.

What is the system reliability defined by any one of four failing?

$$R_s = R_c^4 = 0.999^4 = 99.6\%$$

What is the system availability defined by at least two components working?

$$\begin{aligned} A_s &= \sum_{i=0}^M \frac{N!}{i! (N-i)!} R_c^{N-i} (1 - R_c)^i \\ &= \sum_{i=0}^2 \frac{4!}{i! (4-i)!} 0.999^{4-i} (0.001)^i \\ &= 99.9999996\% \end{aligned}$$



Series and Parallel System Summary

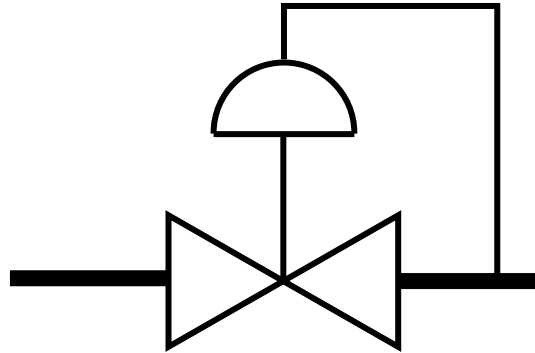
A system of components is less reliable than any of its components

Redundancy increases system availability

Multi-state Systems

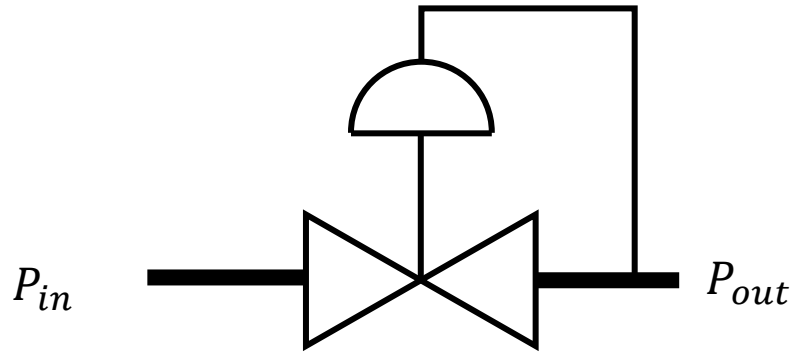
Many systems can fail in different ways, each with different consequences.

A valve can fail closed. A valve can fail open.



Example

A flow control valve maintains constant downstream pressure. Upstream pressure is much greater than desired downstream pressure. If a valve fails closed, flow is blocked. If the valves fail open, downstream pressure will rise. The consequences of high downstream pressure are more serious than no flow. Formulate an expression for the reliability of normal operation if we define failure as no flow or high pressure.

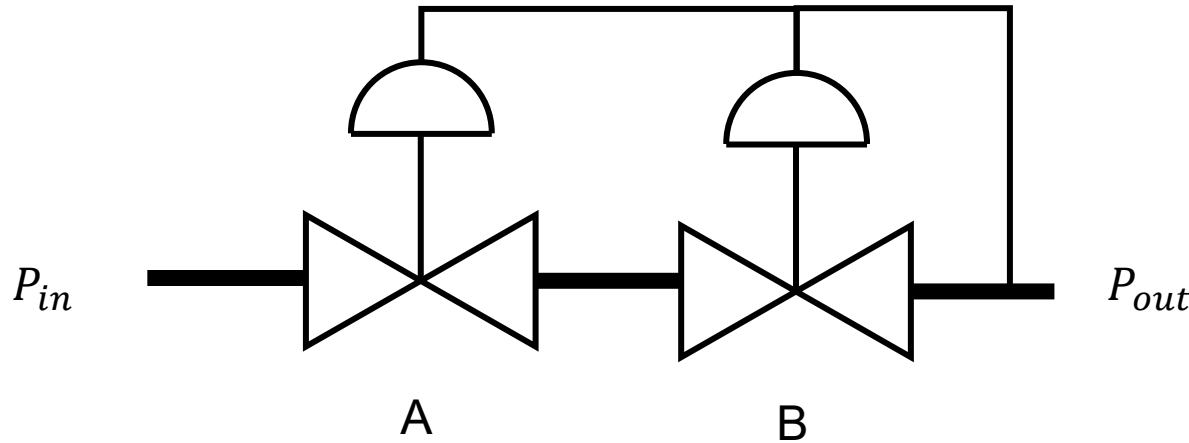


$$R = 1 - (P_c + P_o)$$

State	Probability
Normal	$1 - (P_c + P_o)$
Fail Close	P_c
Fail Open	P_o

Example

Suppose now we have a high severity if we fail open. To prevent this, we can place two valves in series. To analyze this, we first construct a table of all possible system states.

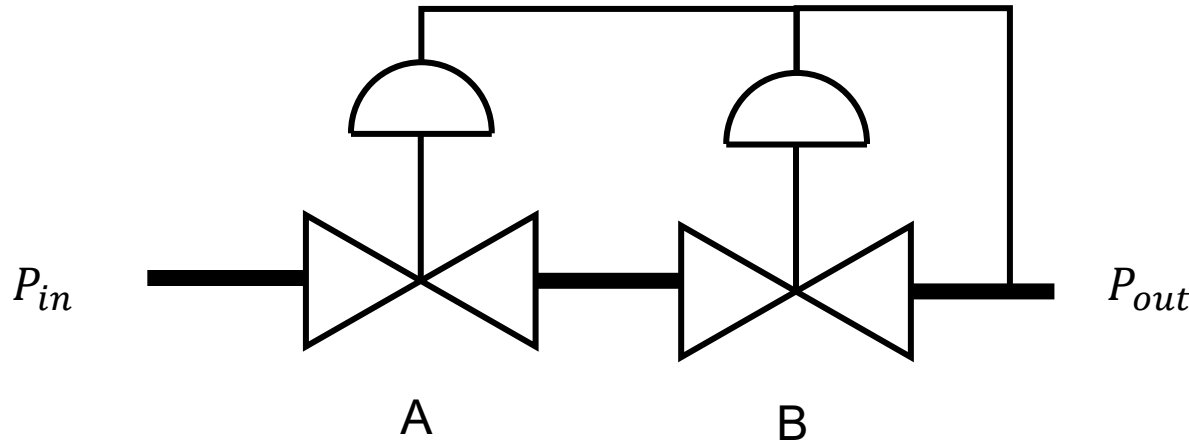


A	B	
Normal	Normal	
Normal	Fail Close	
Normal	Fail Open	
Fail Close	Normal	
Fail Close	Fail Close	
Fail Close	Fail Open	
Fail Open	Normal	
Fail Open	Fail Close	
Fail Open	Fail Open	

Example

Suppose now we have a high severity if we fail open. To prevent this, we can place two valves in series. To analyze this, we must construct a table of all possible system states.

For each state, assess the system consequence.



A	B	Consequence
Normal	Normal	Normal
Normal	Fail Close	Fail Closed
Normal	Fail Open	Normal
Fail Close	Normal	Fail Closed
Fail Close	Fail Close	Fail Closed
Fail Close	Fail Open	Fail Closed
Fail Open	Normal	Normal
Fail Open	Fail Close	Fail Closed
Fail Open	Fail Open	Failed Open

Example

Now assess the probability of being in each state.

A	B	Consequence	Probability
Normal	Normal	Normal	$P_N P_N$
Normal	Fail Close	Fail Closed	$P_N P_C$
Normal	Fail Open	Normal	$P_N P_O$
Fail Close	Normal	Fail Closed	$P_C P_N$
Fail Close	Fail Close	Fail Closed	$P_C P_C$
Fail Close	Fail Open	Fail Closed	$P_C P_O$
Fail Open	Normal	Normal	$P_O P_N$
Fail Open	Fail Close	Fail Closed	$P_O P_C$
Fail Open	Fail Open	Failed Open	$P_O P_O$

Example

Now assess the probability of being in each state.

Add up the reliabilities of each consequence.

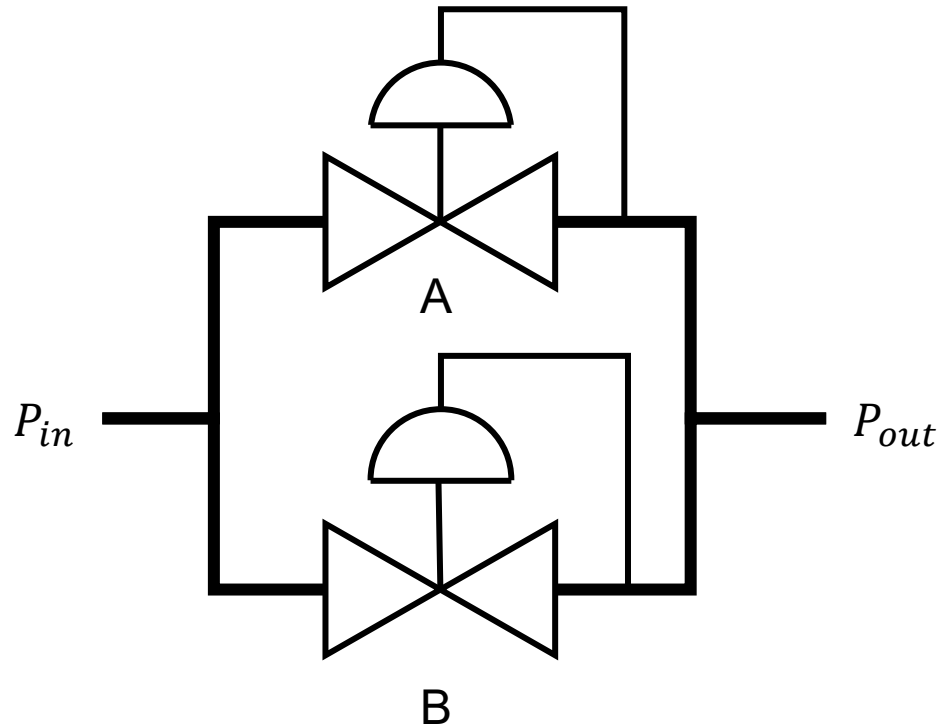
- Probability of Failed Open = P_o^2
- Probability of Failed Closed = $1 - (1 - P_c)^2$
- Probability of Failure = $1 - (1 - P_c)^2 + P_o^2$
- Reliability = $(1 - P_c)^2 - P_o^2$

We have reduced the probability of failing open at the expense of increased probability of failing closed. If $P_o < P_c$, we also reduced the reliability of normal operation.

A	B	Consequence	Probability
Normal	Normal	Normal	$P_N P_N$
Normal	Fail Close	Fail Closed	$P_N P_C$
Normal	Fail Open	Normal	$P_N P_o$
Fail Close	Normal	Fail Closed	$P_C P_N$
Fail Close	Fail Close	Fail Closed	$P_C P_C$
Fail Close	Fail Open	Fail Closed	$P_C P_o$
Fail Open	Normal	Normal	$P_o P_N$
Fail Open	Fail Close	Fail Closed	$P_o P_C$
Fail Open	Fail Open	Failed Open	$P_o P_o$

Example

Suppose now we have a high severity if we fail closed. To prevent this, we can place two valves in parallel. To analyze this, again construct a table of all possible system states.



A	B	Consequence	Probability
Normal	Normal	Normal	$P_N P_N$
Normal	Fail Close	Normal	$P_N P_C$
Normal	Fail Open	Fail Open	$P_N P_O$
Fail Close	Normal	Normal	$P_C P_N$
Fail Close	Fail Close	Fail Close	$P_C P_C$
Fail Close	Fail Open	Fail Open	$P_C P_O$
Fail Open	Normal	Fail Open	$P_O P_N$
Fail Open	Fail Close	Fail Open	$P_O P_C$
Fail Open	Fail Open	Fail Open	$P_O P_O$

Example

Adding up the reliabilities of each consequence.

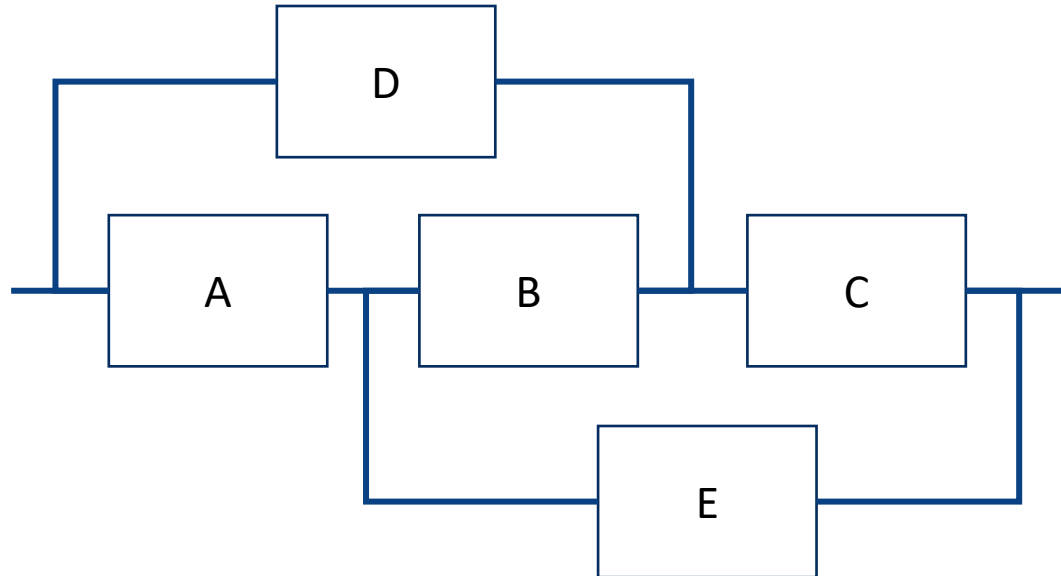
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A	B	Consequence	Probability
Normal	Normal	Normal	$P_N P_N$
Normal	Fail Close	Normal	$P_N P_C$
Normal	Fail Open	Fail Open	$P_N P_o$
Fail Close	Normal	Normal	$P_C P_N$
Fail Close	Fail Close	Fail Close	$P_C P_C$
Fail Close	Fail Open	Fail Open	$P_C P_o$
Fail Open	Normal	Fail Open	$P_o P_N$
Fail Open	Fail Close	Fail Open	$P_o P_C$
Fail Open	Fail Open	Fail Open	$P_o P_o$

Complex Systems

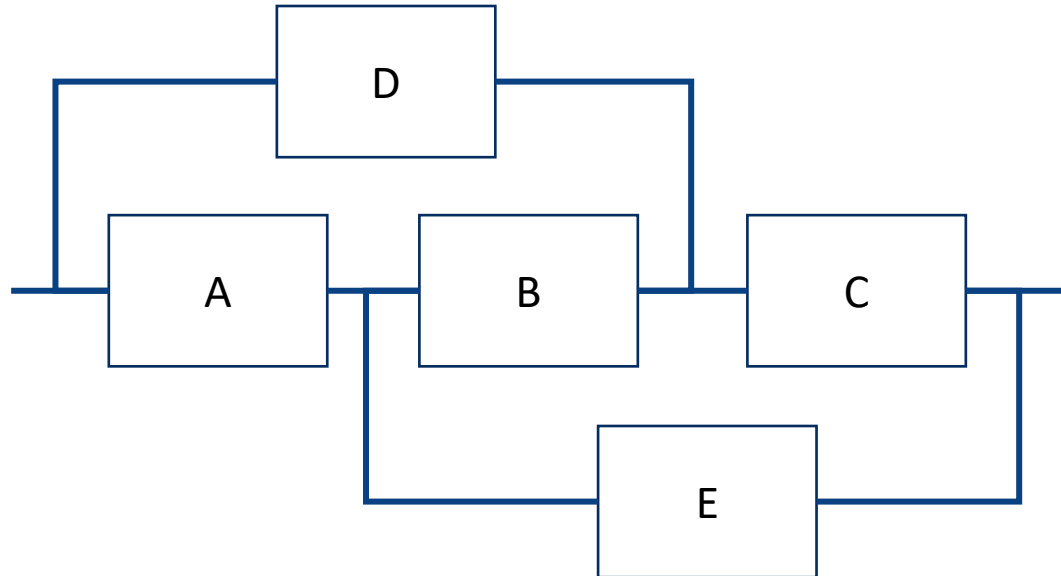
Consider piping, transport, electronic or software systems which have redundant paths.
Here D or (A and B) could be used. E or (B and C) could be used.
Such systems cannot be reduced to a single system element.



Complex System – Example

Define when the system fails in terms of components.

Here, the system works if any of (D C), (A B C), (A E) works.





Complex System – Example

Enumerate the systems states (here as (0,1))

A	B	C	D	E	
1	1	1	1	1	
1	1	1	1	0	
1	1	1	0	1	
1	1	1	0	0	
1	1	0	1	1	
1	1	0	1	0	
1	1	0	0	1	
1	1	0	0	0	
1	0	1	1	1	
1	0	1	1	0	
1	0	1	0	1	
1	0	1	0	0	
1	0	0	1	1	
1	0	0	1	0	
1	0	0	0	1	
1	0	0	0	0	
0	1	1	1	1	
0	1	1	1	0	
0	1	1	0	1	
0	1	1	0	0	
0	1	0	1	1	
0	1	0	1	0	
0	1	0	0	1	
0	1	0	0	0	
0	0	1	1	1	
0	0	1	1	0	
0	0	1	0	1	
0	0	1	0	0	
0	0	0	1	1	
0	0	0	1	0	
0	0	0	0	1	
0	0	0	0	0	



Complex System – Example

Enumerate the systems states (here as (0,1))

Assess when the system works:

if any of (D C), (A B C), (A E) are true:

$$S_{works} = \max(DC, ABC, AE)$$

A	B	C	D	E	System
1	1	1	1	1	1
1	1	1	1	0	1
1	1	1	0	1	1
1	1	1	0	0	1
1	1	0	1	1	1
1	1	0	1	0	0
1	1	0	0	1	1
1	1	0	0	0	0
1	0	1	1	1	1
1	0	1	1	0	1
1	0	1	0	1	1
1	0	1	0	0	0
1	0	0	1	1	1
1	0	0	1	0	0
1	0	0	0	1	1
1	0	0	0	0	0
0	1	1	1	1	1
0	1	1	1	0	1
0	1	1	0	1	0
0	1	1	0	0	0
0	1	0	1	1	0
0	1	0	1	0	0
0	1	0	0	1	0
0	1	0	0	0	0
0	0	1	1	1	1
0	0	1	1	0	1
0	0	1	0	1	0
0	0	1	0	0	0
0	0	0	1	1	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	0



Complex System – Example

Enumerate the systems states (here as (0,1))

Assess when the system works:

if any of (D C), (A B C), (A E) are true:

$$S_{works} = \max(DC, ABC, AE)$$

Suppose $\Pr(\text{fail}) = 0.1$ for any component.

The probability of each row is the product of each component probability:

$$Pr_i = A_i B_i C_i D_i E_i$$

Therefore the probability of each row is:

$$Pr_i = 0.1^{\binom{\# \text{ failed}}{\text{in row } i}} 0.9^{\binom{\# \text{ working}}{\text{in row } i}}$$

A	B	C	D	E	Pr	System
1	1	1	1	1	0.59049	1
1	1	1	1	0	0.06561	1
1	1	1	0	1	0.06561	1
1	1	1	0	0	0.00729	1
1	1	0	1	1	0.06561	1
1	1	0	1	0	0.00729	0
1	1	0	0	1	0.00729	1
1	1	0	0	0	0.00081	0
1	0	1	1	1	0.06561	1
1	0	1	1	0	0.00729	1
1	0	1	0	1	0.00729	1
1	0	1	0	0	0.00081	0
1	0	0	1	1	0.00729	1
1	0	0	1	0	0.00081	0
1	0	0	0	1	0.00081	1
1	0	0	0	0	0.00009	0
0	1	1	1	1	0.06561	1
0	1	1	1	0	0.00729	1
0	1	1	0	1	0.00729	0
0	1	1	0	0	0.00081	0
0	1	0	1	1	0.00729	0
0	1	0	1	0	0.00081	0
0	1	0	0	1	0.00081	0
0	1	0	0	0	0.00009	0
0	0	1	1	1	0.00729	1
0	0	1	1	0	0.00081	1
0	0	1	0	1	0.00081	0
0	0	1	0	0	0.00009	0
0	0	0	1	1	0.00081	0
0	0	0	1	0	0.00009	0
0	0	0	0	1	0.00009	0
0	0	0	0	0	0.00001	0



Complex System – Example

Enumerate the systems states (here as (0,1))

Assess when the system works:

if any of (D C), (A B C), (A E) are true:

$$S_{works} = \max(DC, ABC, AE)$$

Suppose $\Pr(\text{fail}) = 0.1$ for any component.

The probability of each row is:

$$Pr_i = 0.1^{\binom{\# \text{ failed}}{\text{in row } i}} 0.9^{\binom{\# \text{ working}}{\text{in row } i}}$$

Compute the sum of rows where the system is working to determine the system availability.

$$A_s = 0.9712$$

A	B	C	D	E	Pr	System
1	1	1	1	1	0.59049	1
1	1	1	1	0	0.06561	1
1	1	1	0	1	0.06561	1
1	1	1	0	0	0.00729	1
1	1	0	1	1	0.06561	1
1	1	0	1	0	0.00729	0
1	1	0	0	1	0.00729	1
1	1	0	0	0	0.00081	0
1	0	1	1	1	0.06561	1
1	0	1	1	0	0.00729	1
1	0	1	0	1	0.00729	1
1	0	1	0	0	0.00081	0
1	0	0	1	1	0.00729	1
1	0	0	1	0	0.00081	0
1	0	0	0	1	0.00081	1
1	0	0	0	0	0.00009	0
0	1	1	1	1	0.06561	1
0	1	1	1	0	0.00729	1
0	1	1	0	1	0.00729	0
0	1	1	0	0	0.00081	0
0	1	0	1	1	0.00729	0
0	1	0	1	0	0.00081	0
0	1	0	0	1	0.00081	0
0	1	0	0	0	0.00009	0
0	0	1	1	1	0.00729	1
0	0	1	1	0	0.00081	1
0	0	1	0	1	0.00081	0
0	0	1	0	0	0.00009	0
0	0	0	1	1	0.00081	0
0	0	0	1	0	0.00009	0
0	0	0	0	1	0.00009	0
0	0	0	0	0	0.00001	0



Overview

Reliability

Weibull distribution

System Reliability

MTBF

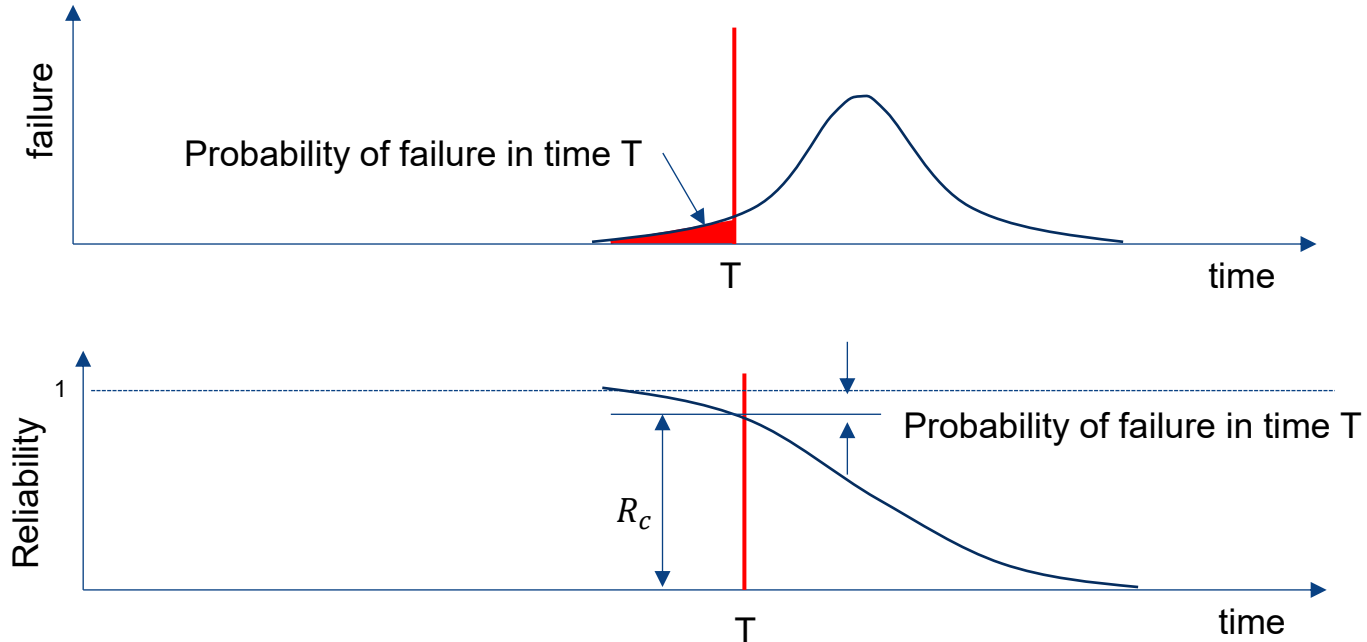
Hazard Rate

Maintenance Plans

Reliability

We defined the **reliability** R_c of a component for a time period T .

It is not the only measure of reliability.

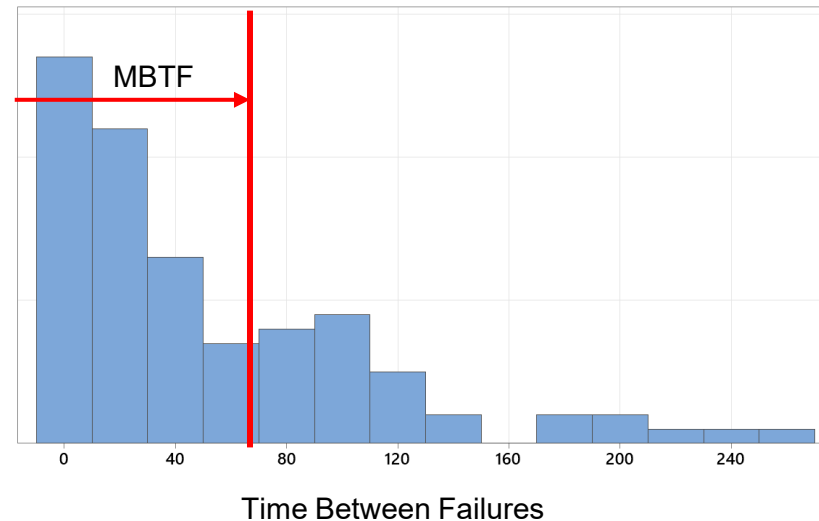
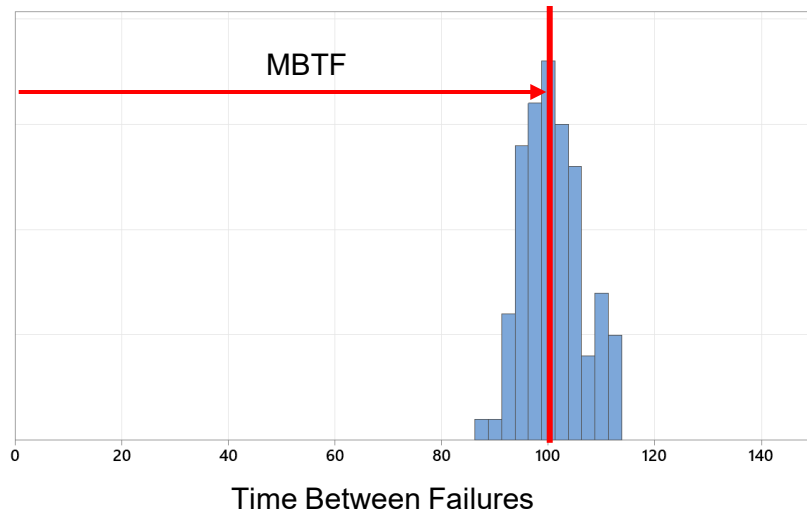


How Do We Measure Reliability?

The time until any one unit fails measures its life.

For a population, the mean time to failure (MTTF) would be a natural measure.

For parts that are replaced, we call it the **mean time between failures (MTBF)**.





Mathematical Relationships

A metric often discussed with reliability is the average life. Consider when a failure is repaired and returned to service. The ***mean-time-between-failures*** is the average time from starting until a failure.

Mathematically, it is the average:

$$MTBF = \int_0^{\infty} R(t)dt$$

And similarly

$$MTBF = \int_{-\infty}^{\infty} tf(t)dt \quad \text{and} \quad MTBF = \int_0^{\infty} (1 - F(t))dt$$



Calculating the Hazard Rate

The ***average hazard rate*** λ is the average number of failures per unit time or actuation.

Over a long duration, the average hazard rate for a part is:

$$\lambda = \frac{1}{MTBF}$$

Calculating the Hazard Rate

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The average hazard rate for a system is sum of the average hazard rates of its elements

$$\lambda_S = \lambda_1 + \lambda_2 + \cdots + \lambda_N$$

Calculating the Hazard Rate

The **average hazard rate** λ is the average number of failures per unit time or actuation. Over a long duration, the average hazard rate for a part is:

$$\lambda = \frac{1}{MTBF}$$

The average hazard rate for a system is sum of the average hazard rates of its elements

$$\lambda_S = \lambda_1 + \lambda_2 + \dots + \lambda_N$$

And so

$$MTBF_S = \left(\frac{1}{MTBF_1} + \frac{1}{MTBF_2} + \dots + \frac{1}{MTBF_N} \right)^{-1}$$



Calculating System MTBF – Example

A system is comprised of four subsystems.

The hazard rates for the subsystems are as follows:

$$\lambda_1 = 1 \text{ failures/10K hours}$$

$$\lambda_2 = 2 \text{ failures/10K hours}$$

$$\lambda_3 = 1 \text{ failures/10K hours}$$

$$\lambda_4 = 5 \text{ failures/10K hours}$$

What's the system MTBF?



Overview

Reliability

Weibull distribution

System Reliability

MTBF

Hazard Rate

Maintenance Plans



Reliability and Cost of Ownership

How much does it cost to own a system?

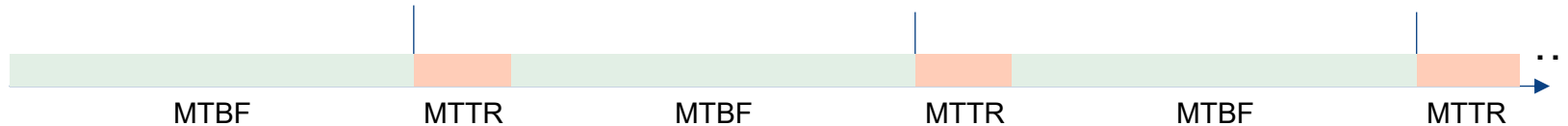
- Purchase price
- Operating cost: fuel/electricity and consumables
- Downtime: lost profits when not in service
- Service: replacement material and labor cost
- Useful life: value during life
- Residual value: costs or depreciation upon end of life

The sum of these costs is the ***total cost of ownership***.

Service Rate

Consider averages for a system, the ***mean time between failures*** (MTBF) and the ***mean time to repair*** (MTTR). The sum of MTBF and MTTR is the full time to the next failure.

On average, total life is the sum of these over a period of service and repair cycles.



In-service and down-time ratios are fractions of total time

$$\text{In Service Rate} = \frac{MTBF}{MTBF + MTTR}$$

$$\text{Downtime Rate} = \frac{MTTR}{MTBF + MTTR}$$

How MTBF affects Service and Downtime

Suppose a system has a reliability of 1000 hrs MTBF

Suppose the service labor is 25 hrs MTTR

Improvements are made to the system increasing reliability to 1500 hrs MTBF

What is the expected service and downtime rates after the reliability improvements?

$$\text{In Service Rate} = \frac{MTBF}{MTBF + MTTR} = \frac{1000}{1000 + 25} = 97.6\%$$

$$\text{In Service Rate} = \frac{MTBF}{MTBF + MTTR} = \frac{1500}{1500 + 25} = 98.4\%$$

Increasing MTBF increases in-service rate.

Over 10,000 hrs this is 80 hrs more service, and 3 less repair charges.



Service Parts Costs

When a part is replaced, it costs both labor and material.

The average rate of this cost is the ***Service Parts Cost rate***,

$$r_{part} = \frac{C}{MTBF + MTTR}$$

where C is the replacement part cost (labor + material).

If this is desired to be expressed as a cost over a (long) time interval T , then

$$C_{part \text{ per } T} = \frac{CT}{MTBF + MTTR}$$



Service Strategy

When do you shut the system down and repair it?

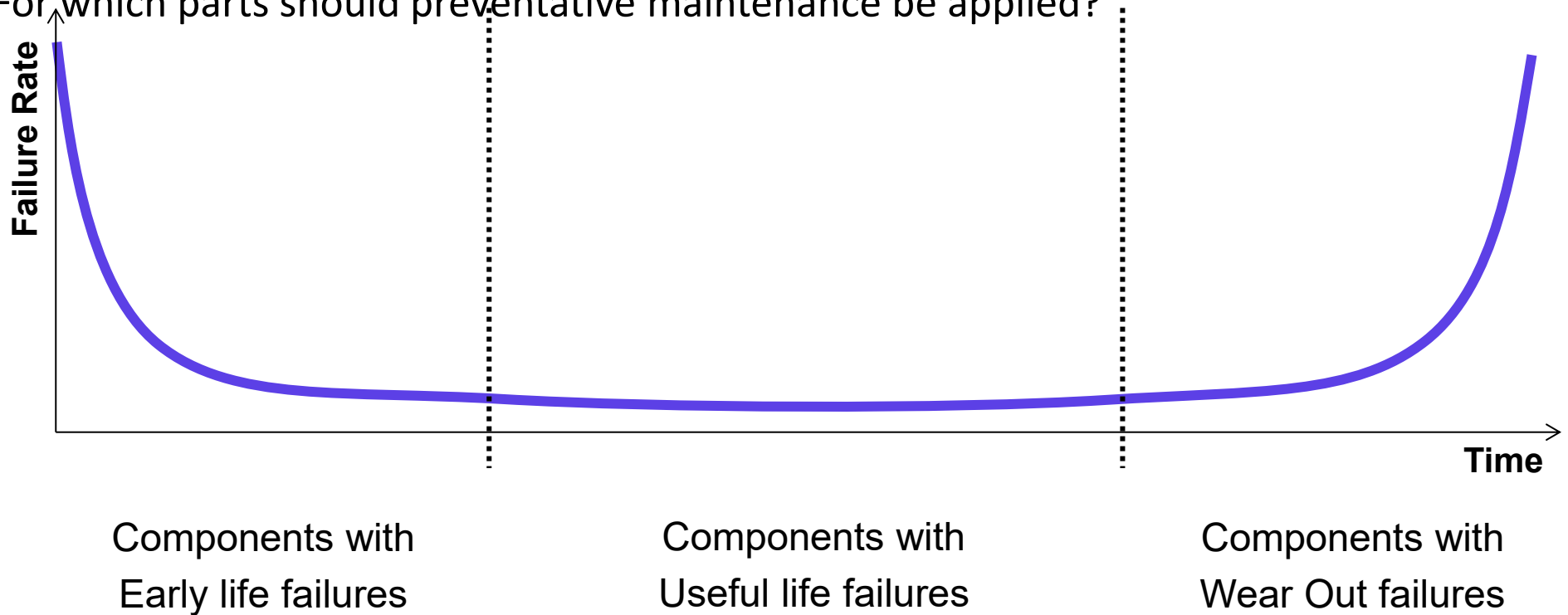
1. Run to Failure
2. Preventive Maintenance

Preventative maintenance replaces good components (just) before they have failed

- Prevents unexpected downtimes of the system
- Enables system to be operational at times needed and scheduled
- Enables repairs during off-hours

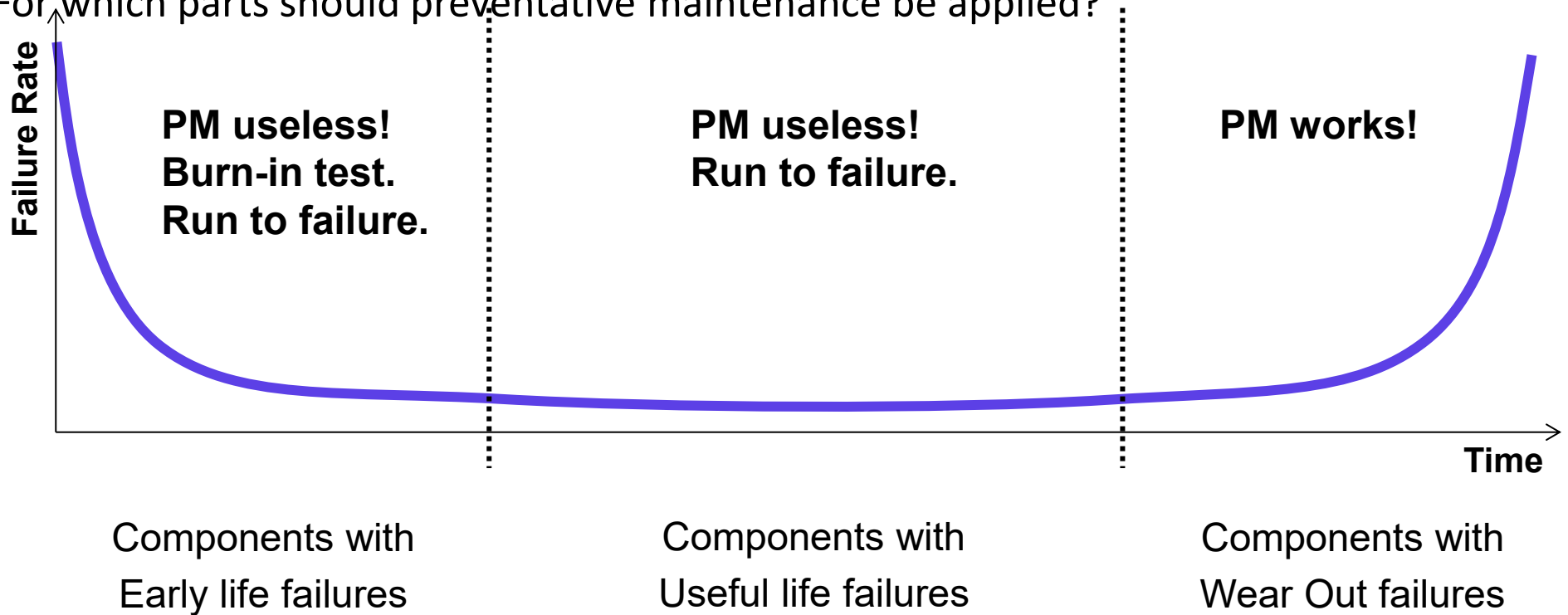
Preventative Maintenance

A system has failures due to many failure modes on many parts.
For which parts should preventative maintenance be applied?



Preventative Maintenance

A system has failures due to many failure modes on many parts.
For which parts should preventative maintenance be applied?





Maintenance Summary

A Good Maintenance Strategy can reduce total service cost

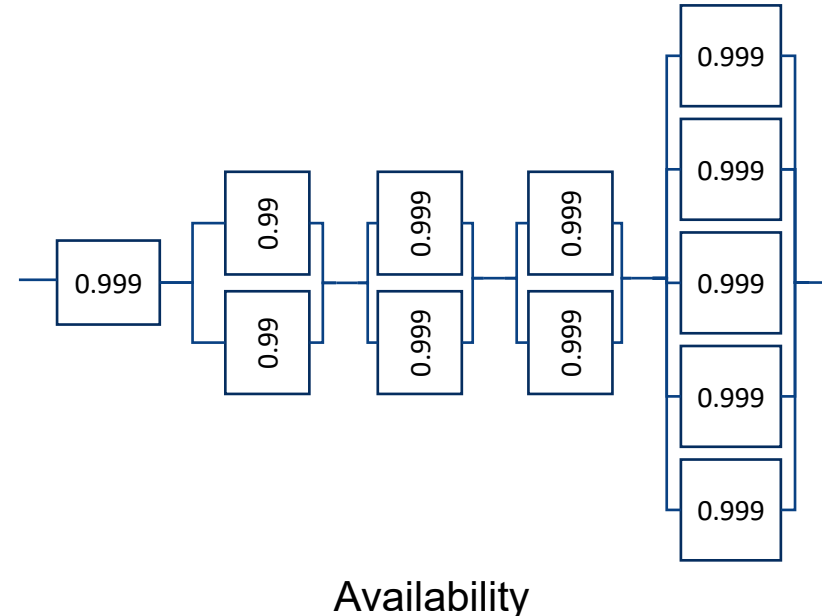
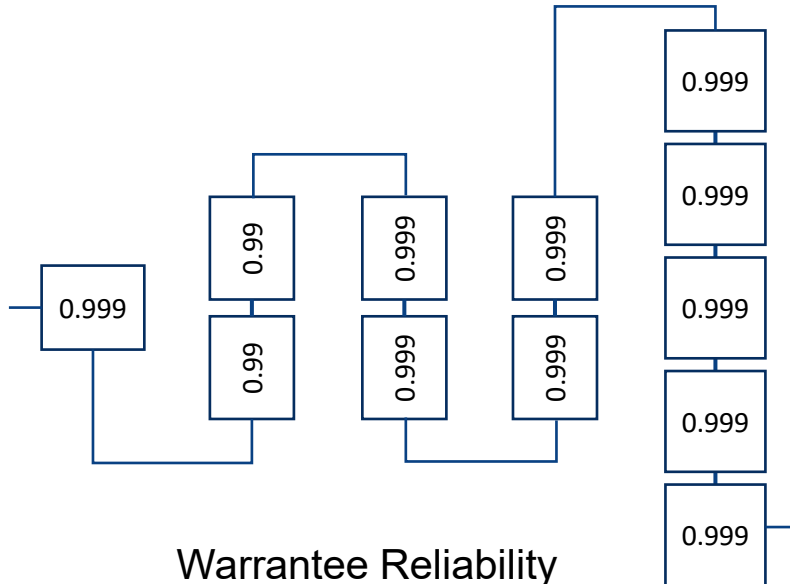
PM doesn't work for failure modes with a decreasing or constant hazard rate

Additional Potential benefits of PM strategy

1. Increased equipment availability for customer
2. Increased customer satisfaction
3. Increased productivity of service force
4. Increased parts sales

Warranty Exercise

Supposing the failure rates for the warranty period are as shown, what is the probability of a service call during the warranty period? What is the availability during the warranty period?



Exercise – Hydraulic Subsystem

What is the system failure rate?

Which component has the biggest share of unreliability?

Which improvement will have the greatest impact on system reliability, a control valve with MTBF=10,000 hrs. or a pump with 750 hours?

Component	MTBF (hrs)	λ (failures/hr)	New MTBF	Reduction
Pump	600	1.66×10^{-3}		
Control valve	3000	3.33×10^{-4}		
Relief valve	2000	5.00×10^{-4}		
PLC Controller	12,000	8.33×10^{-5}		
System				

Exercise – System Service Parts Cost Rate

A system has 5 components with MTBF and replacement costs shown. With negligible MTTR, then what is the service parts cost per 1000 hours?

Component	MTBF (hrs)	Replacement Cost
A	600	\$50
B	3000	\$80
C	2000	\$20
D	12000	\$220
E	200	\$20

Exercise: System Reliability Allocation

Determine the system reliability of each assembly and the overall system.

If you could increase the reliability of only one component, which would you pick to have most impact on improving system reliability?

Compressor			5 Year Reliability		
			R_{Ci}	R_{Ai}	R_{system}
	Motor				
		Contacts	0.99950		
		Thermal cutout	0.99900		
		Bearings	0.99990		
		Armature	0.99999		
		Windings	0.99999		
		Brushes	0.99000		
	Pump				
		Flex plate	0.99990		
		Bearings	0.99990		
		Rotor blades	0.99990		
		Housing	0.99999		
	Controller				
		Breaker	0.99990		
		Switch	0.99990		
		Panel	0.99999		
SYSTEM					

Exercise: System Reliability Allocation

The low reliability items define the system reliability.

Work to redesign them for higher reliability.

Compressor			5 Year Reliability		
			R_{Ci}	R_{Ai}	R_{system}
	Motor		0.9884		
		Contacts	0.99950		
		Thermal cutout	0.99900		
		Bearings	0.99990		
		Armature	0.99999		
		Windings	0.99999		
		Brushes	0.99000		
	Pump		0.9998		
		Flex plate	0.99990		
		Bearings	0.99990		
		Rotor blades	0.99990		
		Housing	0.99999		
	Controller		0.9998		
		Breaker	0.99990		
		Switch	0.99990		
		Panel	0.99999		
SYSTEM			0.9880		



Service Strategy

Every part will at some point fail.

What is your strategy for getting it fixed for the customer?

1. Toss out the system and replace with a new one.
2. Part repair. Disassemble the unit to the part, replace the part, and reassemble.
3. Module repair. Bundle the part into a module called the *replacement unit*.
Disassemble to the RU, replace the RU, reassemble.

Look at

- 1) Difficulty: how hard it is to disassemble to the part/RU (number of steps),
how precise must the re-assembly must be,
- 2) MBTF: how often must the part/RU be repaired?



Replacement Modules

The service strategy is designed. ***What parts are bundled into each replaceable unit?***

Consider

- The costs of the parts in the RU.
- The life of the parts in the RU.
- The ease of disassembly and assembly of the RU. Field replaceable?
- The end-of-life decision for the RU: field repair, refurbish, recycle, dispose?

All of these are dependent on the replaceable unit modularity, which in turn impacts the service strategy and total cost of ownership.

End of Life Alternatives

Once it fails, what do you do?

Customer Repair	Service Repair	Recycle	Landfill
			

Refill, no waste.

Ideally remanufacture the failed part back to new.

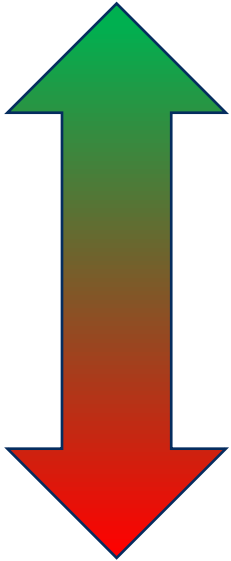
Ideally the replacement module is all one material and doesn't need disassembly.

This is what happens when end of life is not engineered.

End of Life Circularity Strategy

How did you design for the end-of-life of each replaceable unit?

Most
sustainable



Least
sustainable

Field Repair	Enable the customer to fix the RU and keep using it. No failed parts to manage.
Field replace, remote refurbish	Customer removes and replaces RU, and sends back the failed RU to special handlers where it is refurbished and returned to inventory and reused.
Field replace, field recycle	Customer removes and replaces the RU, and the failed RU is “pure” material and can be put whole into municipal recycling for the material.
Field replace, remote recycle	Customer removes and replaces the RU, and the failed RU is “pure” material and special handlers can grind /melt it and recycle the material.
Field replace, field separate and recycle	Customer removes and replaces the RU, and the failed RU can be easily separated and each part put in separate municipal recycle bins for the materials.
Field replace, remote separate and recycle	Customer removes and replaces the RU, and the failed RU is sent to special handlers who can separate parts and recycle their materials.
Field replace, remote separate, recycle and dispose portions	Customer removes and replaces the RU, and the failed RU is sent to special handlers who can separate parts and recycle some materials, yet some materials can only be disposed.
Dispose	Customer removes and replaces the RU and must dispose of the failed RU.

Service Strategy

With a new design, you can isolate or combine components for servicing.

Make an indented list of components by assembly.

Roll up reliabilities as a series system to consider service and repair.

Compressor			Reliability		
			R_{Ci}	R_{Ai}	R_{system}
	Motor			0.9884	
		Contacts	0.99950		
		Thermal cutout	0.99900		
		Bearings	0.99990		
		Armature	0.99999		
		Windings	0.99999		
		Brushes	0.99000		
	Pump			0.9998	
		Flex plate	0.99990		
		Bearings	0.99990		
		Rotor blades	0.99990		
		Housing	0.99999		
	Controller			0.9998	
		Breaker	0.99990		
		Switch	0.99990		
		Panel	0.99999		
SYSTEM					0.9880

Service Plans

To plan a servicing strategy, *look at the low reliability items*.

Isolate all very low reliability items and make them easily field replaceable.

Also work to increase their reliability.

Compressor			Reliability		
			R_{Ci}	R_{Ai}	R_{system}
	Motor			0.9884	
		Contacts	0.99950		
		Thermal cutout	0.99900		
		Bearings	0.99990		
		Armature	0.99999		
		Windings	0.99999		
		Brushes	0.99000		
	Pump			0.9998	
		Flex plate	0.99990		
		Bearings	0.99990		
		Rotor blades	0.99990		
		Housing	0.99999		
	Controller			0.9998	
		Breaker	0.99990		
		Switch	0.99990		
		Panel	0.99999		
SYSTEM					0.9880

Service Plans

To plan a servicing strategy, *look at the low reliability items.*

For the next lowest reliability units, consider how they might be packaged into replacement modules.

1. Failure mode type
2. Failure rate

Compressor			Reliability			Failure Mode
			R_{Ci}	R_{Ai}	R_{system}	
	Motor			0.9884		
		Contacts	0.99950			Wear
		Thermal cutout	0.99900			Random
		Bearings	0.99990			Wear
		Armature	0.99999			Random
		Windings	0.99999			Random
		Brushes	0.99000			Wear
	Pump			0.9998		
		Flex plate	0.99990			Fatigue
		Bearings	0.99990			Wear
		Rotor blades	0.99990			Pitting, abrasion
		Housing	0.99999			Pitting, abrasion
	Controller			0.9998		
		Breaker	0.99990			Wear
		Switch	0.99990			Wear
		Panel	0.99999			Random
SYSTEM					0.9880	

Service Plans

Suppose you bundle these two items into a wire bundle replaceable unit.

- They are easily removed as a module.
- They are easily replaced as a module.
- No precision tools needed.
- If failed, neither can be repaired, they must be replaced and recycled.

What is the reliability of this service module?

When would this be a bad decision?

When would this be a good decision?

,

Compressor		Reliability			Failure Mode
		R_{Ci}	R_{Ai}	R_{system}	
	Motor		0.9884		
	Contacts	0.99950			Wear
	Thermal cutout	0.99900			Random
	Bearings	0.99990			Wear
	Armature	0.99999			Random
	Windings	0.99999			Random
	Brushes	0.99000			Wear
	Pump		0.9998		
	Flex plate	0.99990			Fatigue
	Bearings	0.99990			Wear
	Rotor blades	0.99990			Pitting, abrasion
	Housing	0.99999			Pitting, abrasion
	Controller		0.9998		
	Breaker	0.99990			Wear
	Switch	0.99990			Wear
	Panel	0.99999			Random
SYSTEM				0.9880	

Service Plans

Change the RU design so that

- The failure modes are the same type
- The higher failure rate parts cost more than the lower failure rate parts



Thermal cutout
Random failure from
1st over heat condition



Thermal switch
Fatigue failure from
multiple over-heats

Compressor		Reliability			Failure Mode
		R_{Ci}	R_{Ai}	R_{system}	
	Motor		0.9879		
	Contacts	0.99950			Wear
	Thermal switch	0.99990			Fatigue
	Bearings	0.99990			Wear
	Armature	0.99999			Random
	Windings	0.99999			Random
	Brushes	0.99000			Wear
	Pump		0.9998		
	Flex plate	0.99990			Fatigue
	Bearings	0.99990			Wear
	Rotor blades	0.99990			Pitting, abrasion
	Housing	0.99999			Pitting, abrasion
	Controller		0.9998		
	Breaker	0.99990			Wear
	Switch	0.99990			Wear
	Panel	0.99999			Random
SYSTEM				0.9875	

Service Plans

Isolate low reliability parts

Bundle wearout parts such that longer life parts in the RU cost less than lower life parts in the RU

Bundle so the assembly/disassembly gets easier

Compressor		Reliability			Failure Mode
		R_{Ci}	R_{Ai}	R_{system}	
	Motor		0.9879		
	Contacts	0.99950	Field repair		Wear
	Thermal switch	0.99990			Fatigue
	Bearings	0.99990			Wear
	Armature	0.99999	Return refurbish		Random
	Windings	0.99999			Random
	Brushes	0.99000	Field repair		Wear
	Pump				
	Flex plate	0.99990			Fatigue
	Bearings	0.99990	Return refurbish		Wear
	Rotor blades	0.99990			Cracking, abrasion
	Housing	0.99999			Pitting, abrasion
	Controller		0.9998		
	Breaker	0.99990			Wear
	Switch	0.99990	Return refurbish		Wear
	Panel	0.99999			Random
SYSTEM				0.9875	



Summary

The reliability of a component is considered by each failure mode individually

The reliability of a system is the rollup sum of the individual failure modes

The Weibull distribution can well represent many failure modes and types

Reliability block diagrams are used to form truth tables to assess availability

The mean time to failure and hazard rate can also be used to represent reliability

A reliability requirement involves a probability of failure at a selected time under load

Maintenance plans require understand of failure mode types and probabilities

NEXT: Distribution Fitting